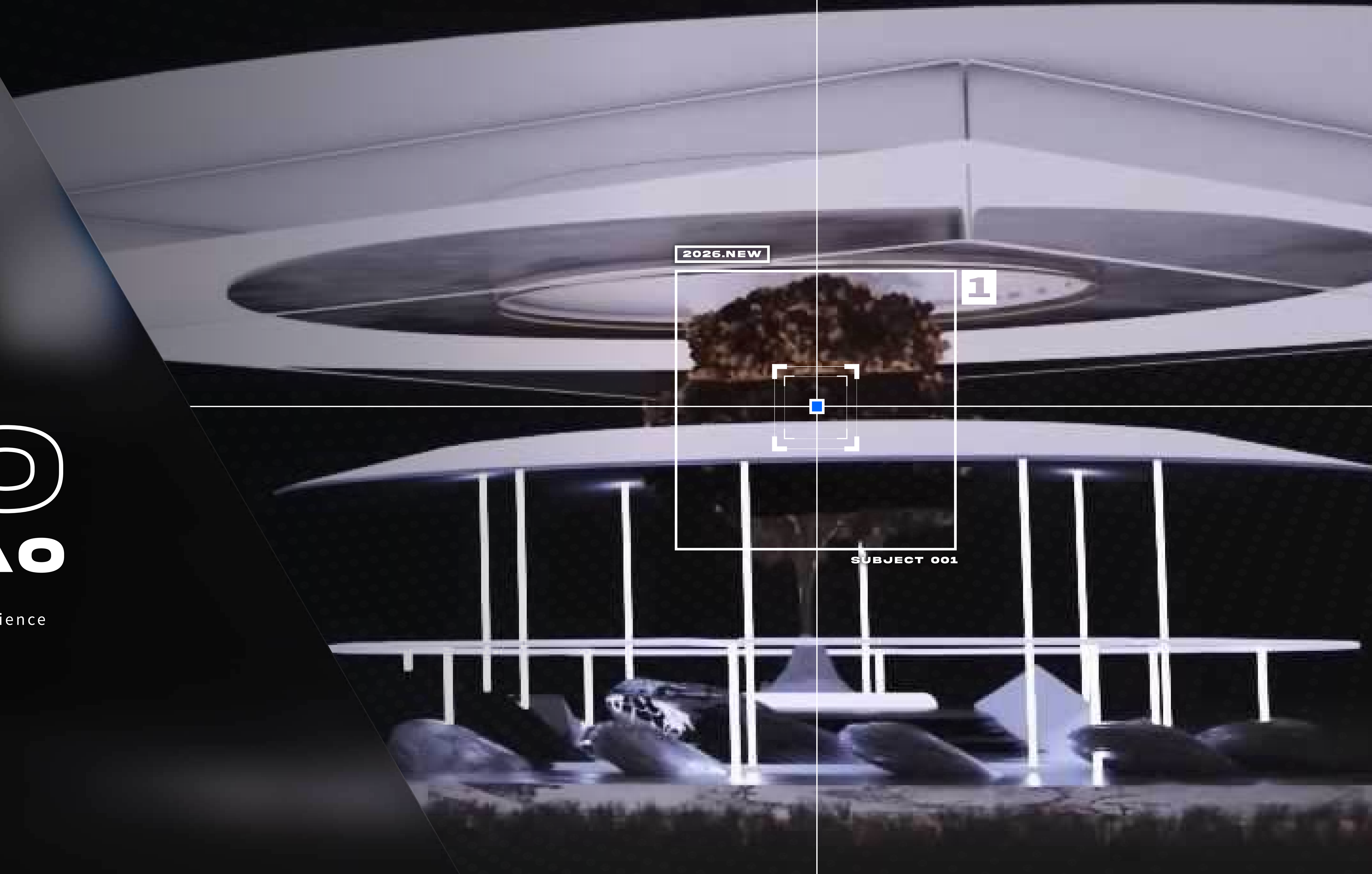
EST. 2026

DESIGN PORTFOLIO

XIAN-HAO (HARRY) LIAO

Harry Liao excels at bridging creative strategy with sci-fi technology, possessing extensive experience in UX design, visual design, 3D modeling, motion graphics, and web development.



DESIGN PORTFOLIO

INTRO

Hello,
I am Xian-Hao Liao 廖先皓, a UI designer with a passion for futuristic aesthetics, storytelling, and immersive experiences.

EXPERIENCE

2025 – 2026	GATE.COM
2025	BOSTON UNIVERSITY
2024-2025	THE UPS STORE
2023	BJ'S CLUB (INTERN)
2022 – 2024	MASSART
2021 – 2022	ABC DESIGN
2016 – 2020	NTCUST

SKILL

Adobe Illustrator
Adobe Photoshop
Adobe After Effects
Figma
Cinema 4D
HTML & CSS & JS
Nano Banana Pro

CONTACT

HOMEPAGE	HARRYLIAO.DESIGN
EMAIL	HARRYXLIAO@GMAIL.COM
WECHAT	HARRYXLIAO

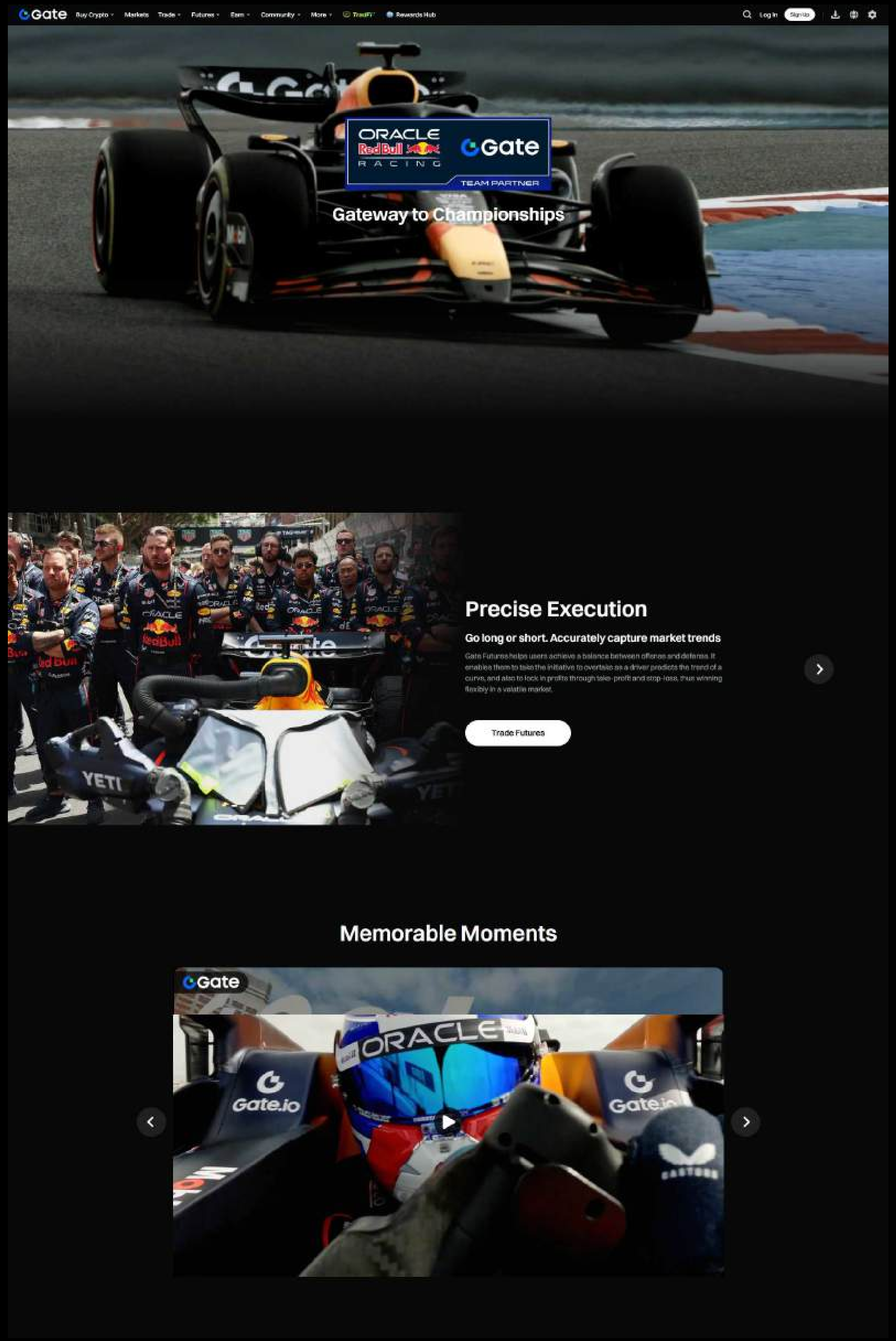
Gate

- Executed experience-focused UX design for the RedBull x Gate partners page and events pages, ensuring high delivery standards.
- Accomplished social motion graphics for four high-quality ad campaigns.

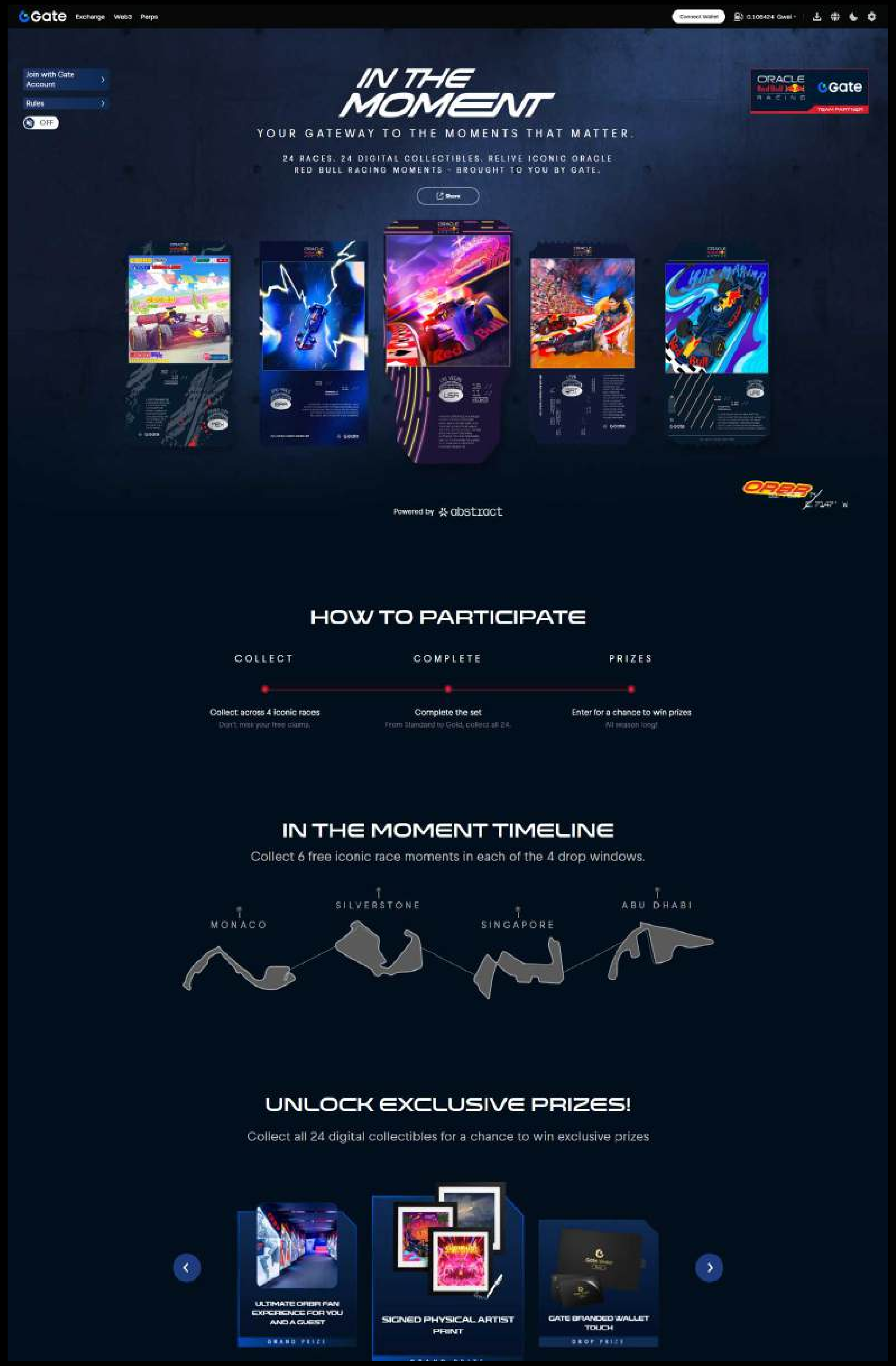
PROJECT SCOPE

UX/UI DESIGN SOCIAL MEDIA VISUALS MOTION GRAPHICS ICON DESIGN

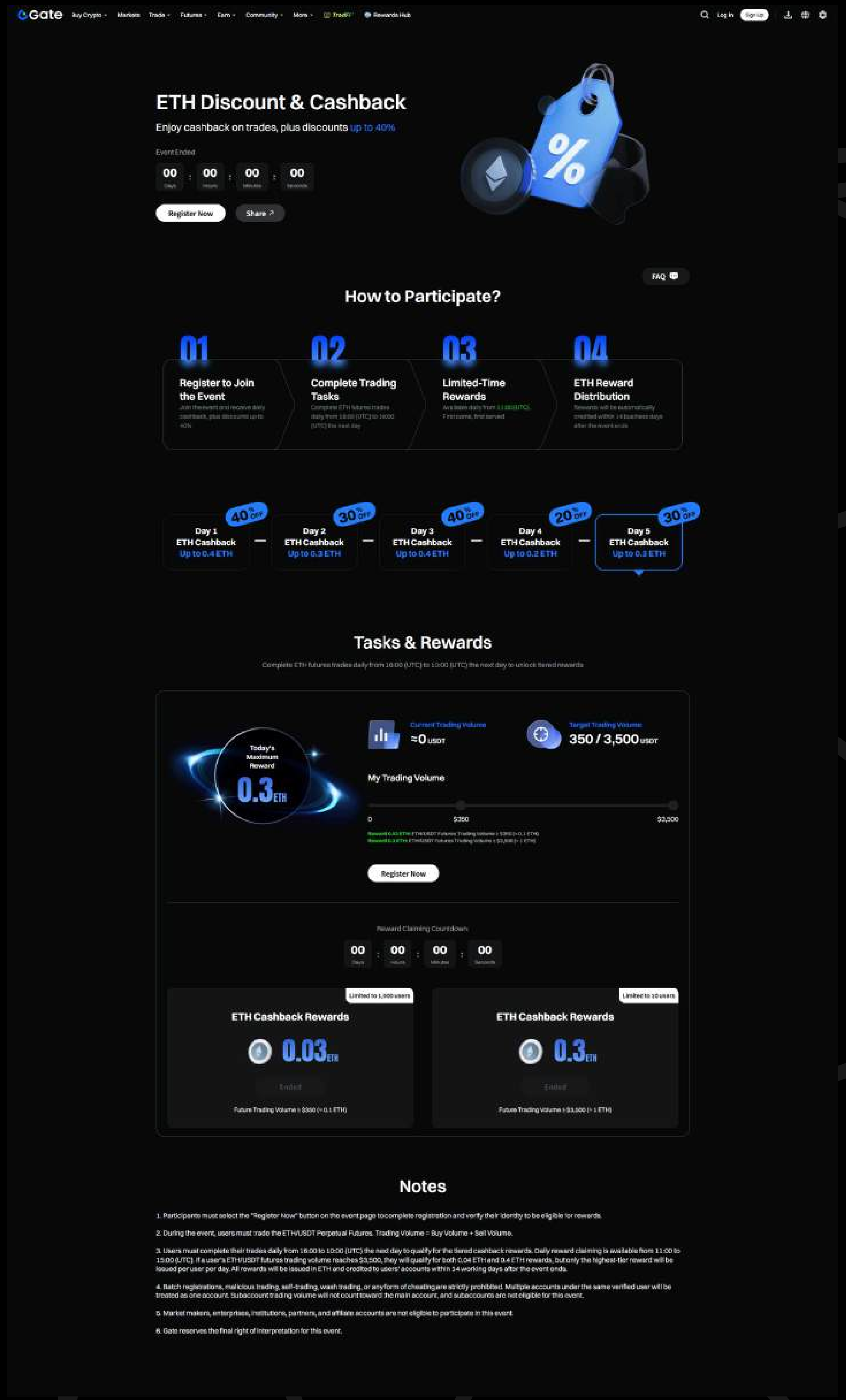
'Gate x Redbull Racing' Partners Page Design



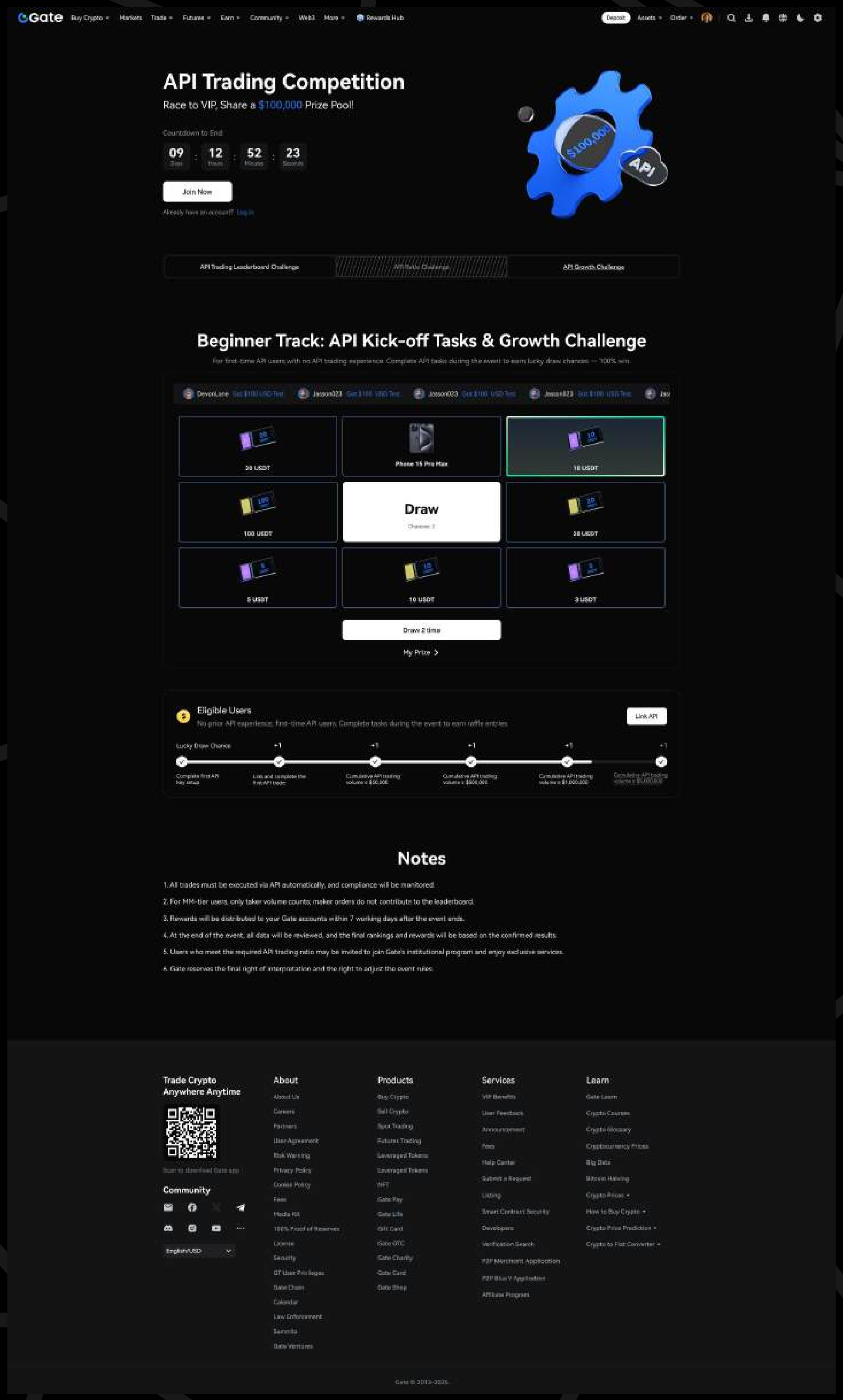
'Gate x Redbull Racing' NFT Website Design



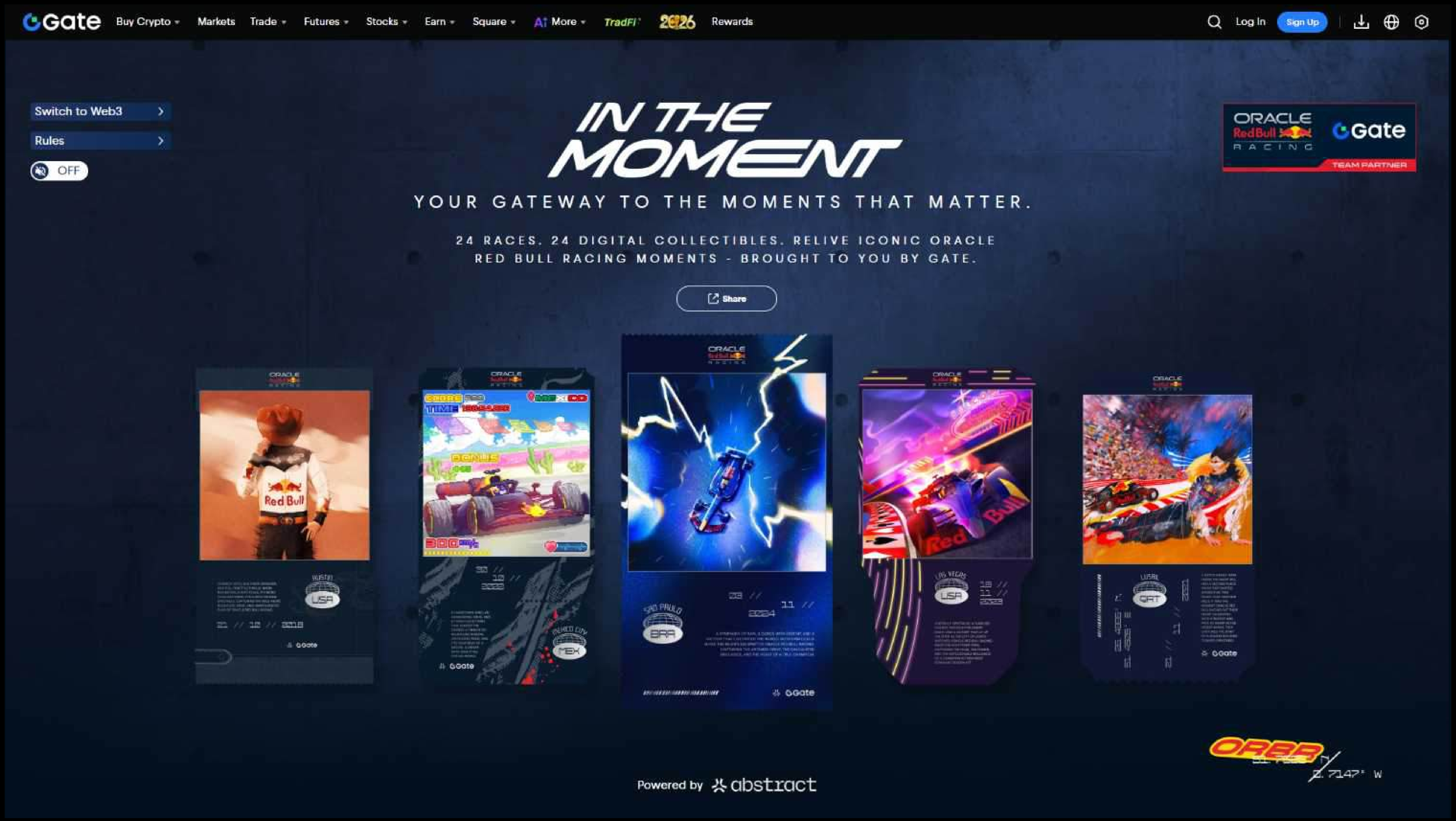
ETH Event Page Design

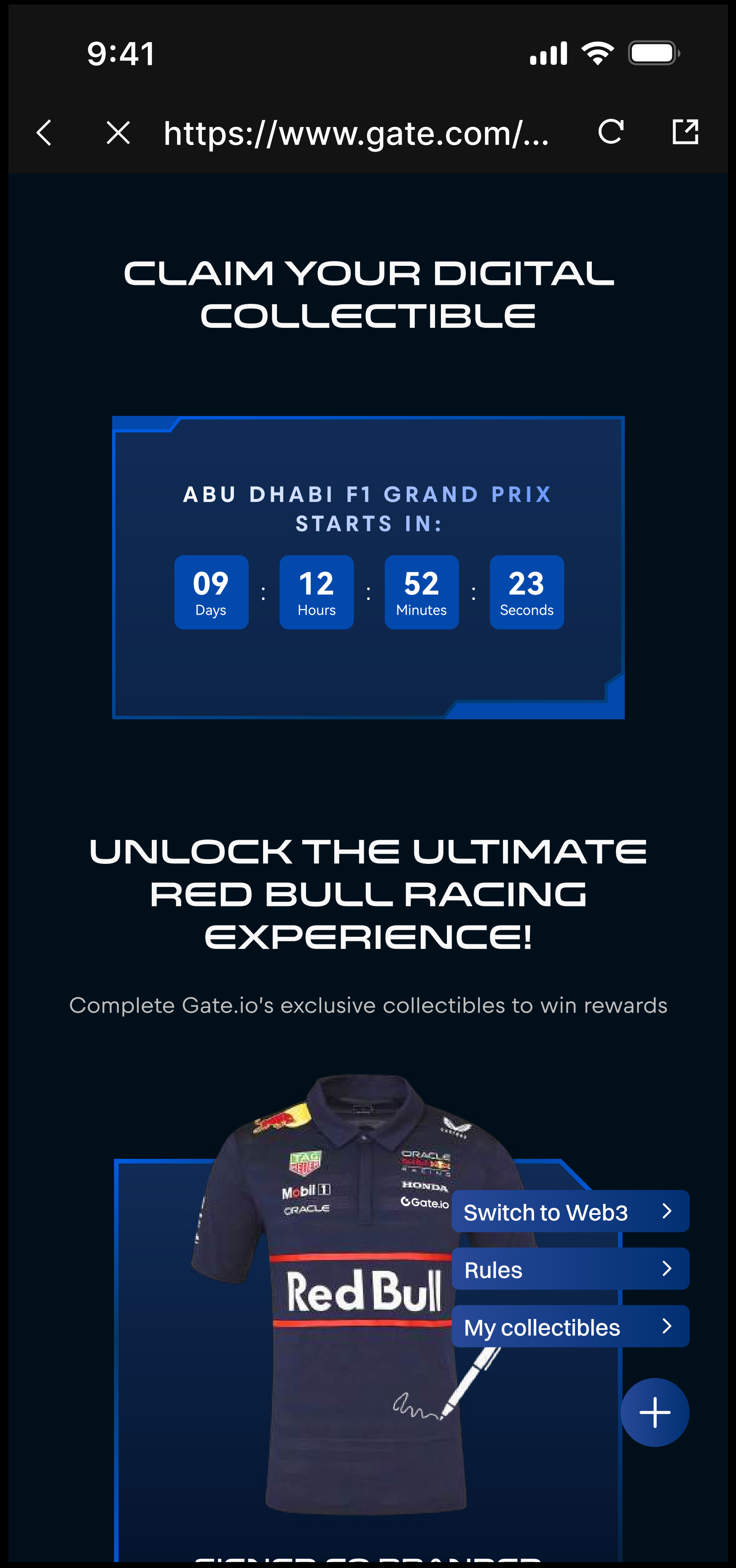


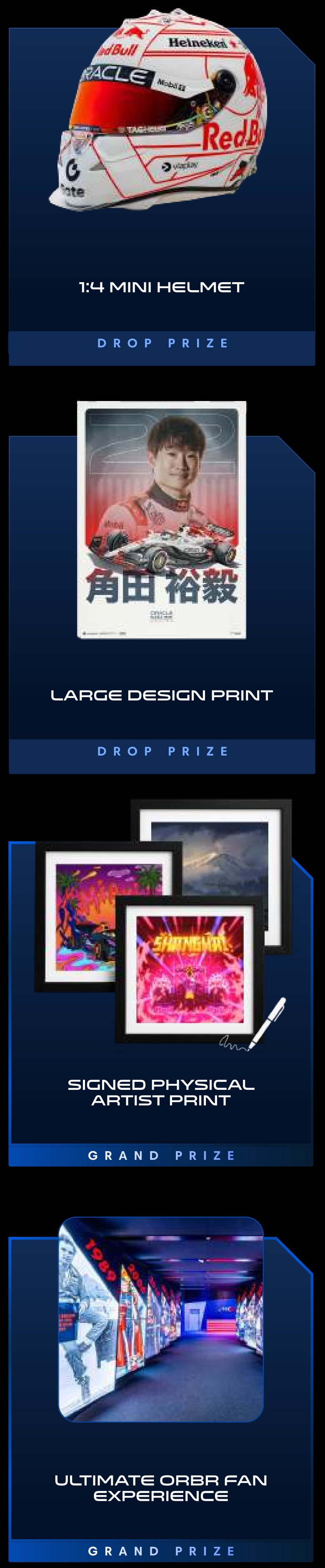
API Trading Page Design



EVENT Website Design



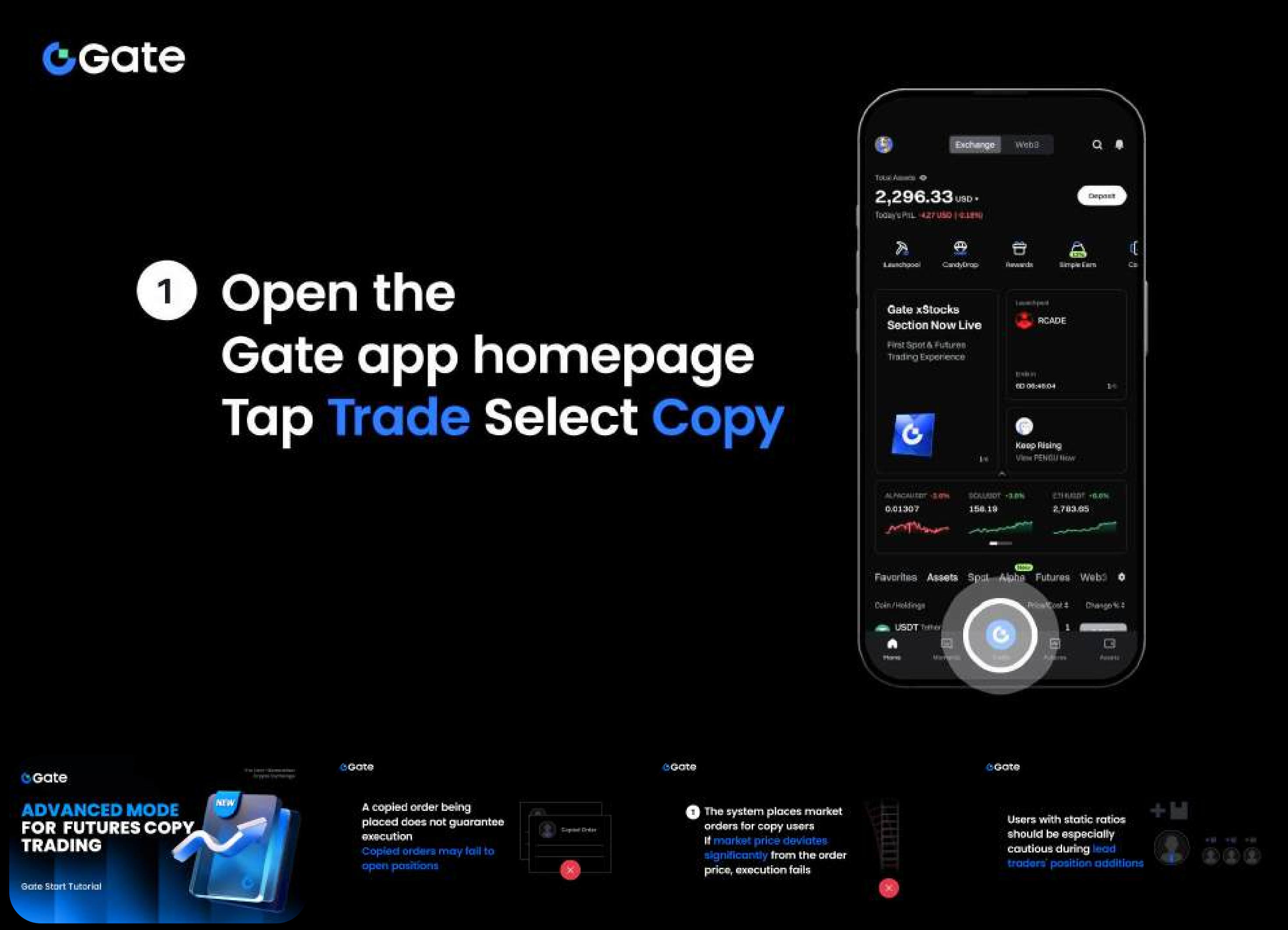


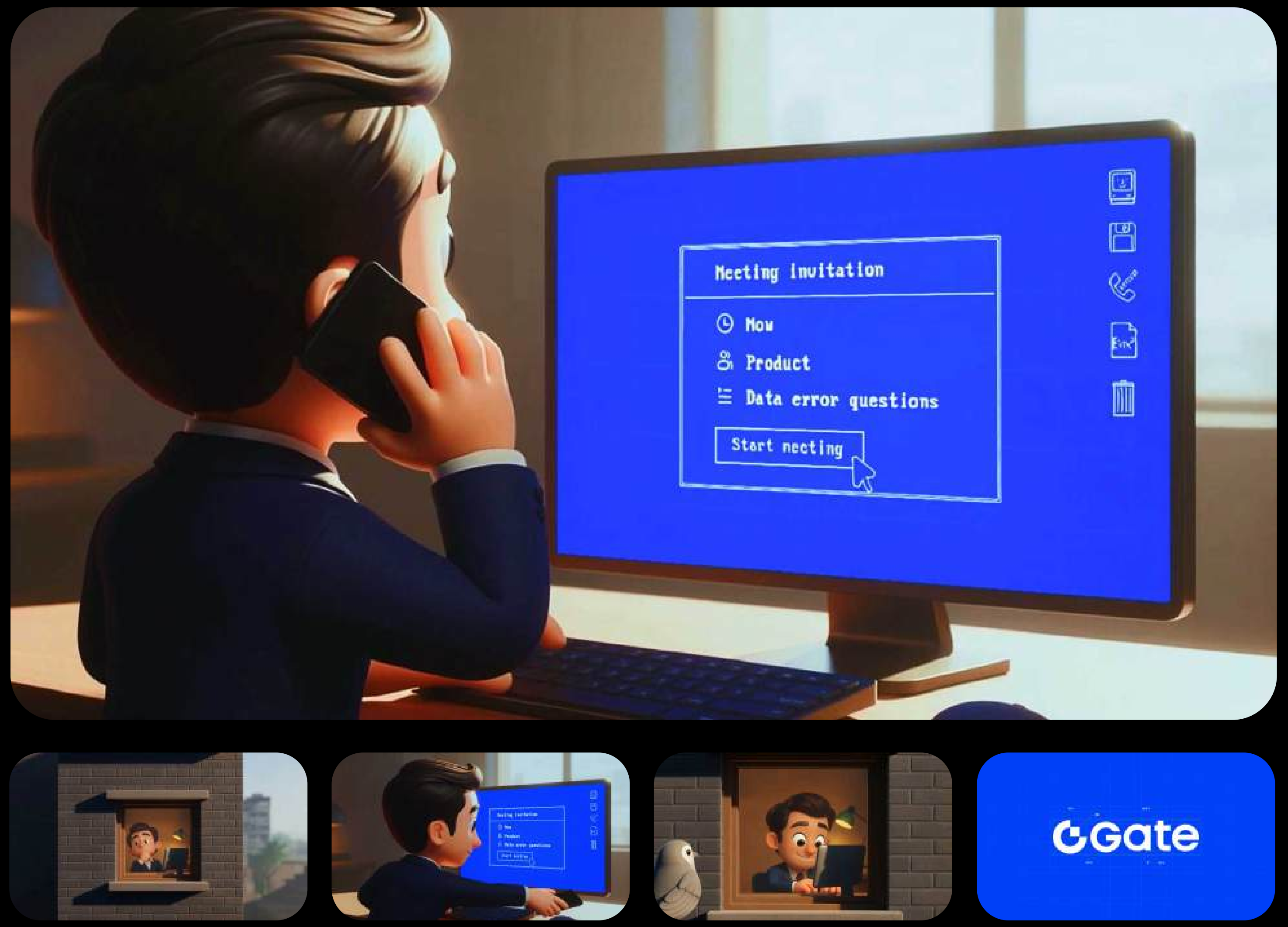


PROJECT TYPE
UX/UI DESIGN

PROJECT SCOPE
GATE X REDBULL RACING NFT EVENT WEBSITE DESIGN

Motion Graphic Design





GATE

3D ICON DESIGN

Collaborated closely with the team to craft dynamic 3D icons for Gate.com event campaigns. These designs powered the V3.0 UI ecosystem prior to the system upgrades.

WEBSITE DOMAIN

GATE.COM

GATE

EVENT ICON DESIGN

In collaboration with the team, I designed custom 3D icons tailored for various Web3 event pages.

EVENT ICON SYSTEM LIST

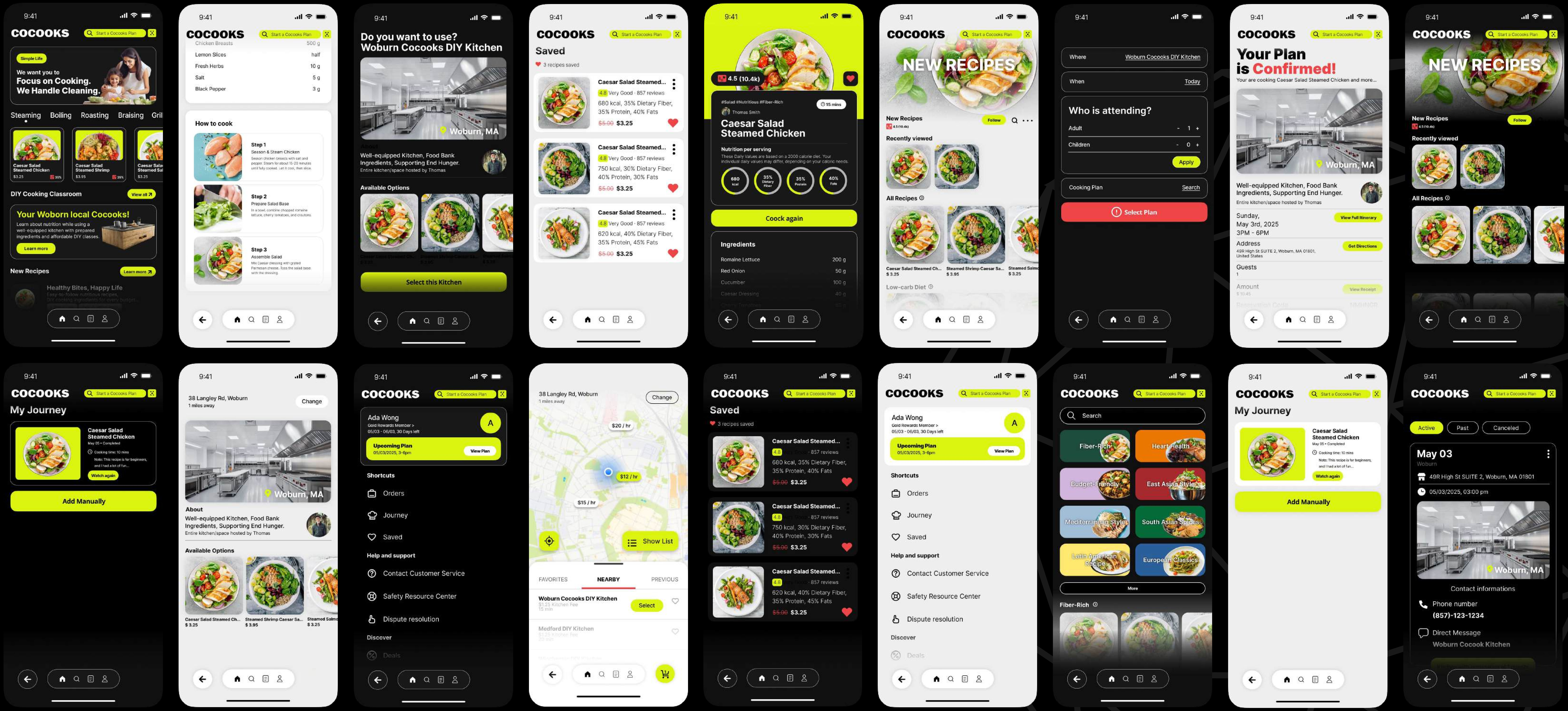
- API TRADING EVENT
- CALL/PUT OPTION BETA
- TRADING BOT REPORT
- ROLLING SHORT OPTIONS
- REWARDS & ACHIEVEMENTS
- CRYPTO DISCOUNT EVENT
- ANNIVERSARY CELEBRATIONS
- TRAVEL & GLOBAL ECOSYSTEM
- CRYPTO WALLETS & ASSETS

COCOOKS

- Designed a mobile app empowering young adults to combat poor nutrition through accessible education and interactive meal preparation.

PROJECT SCOPE
UX/UI DESIGN, APP PROTOTYPING, USER RESEARCH, VISUAL IDENTITY

App UX Design



COCOOKS
INTEGRATED SERVICE DESIGN

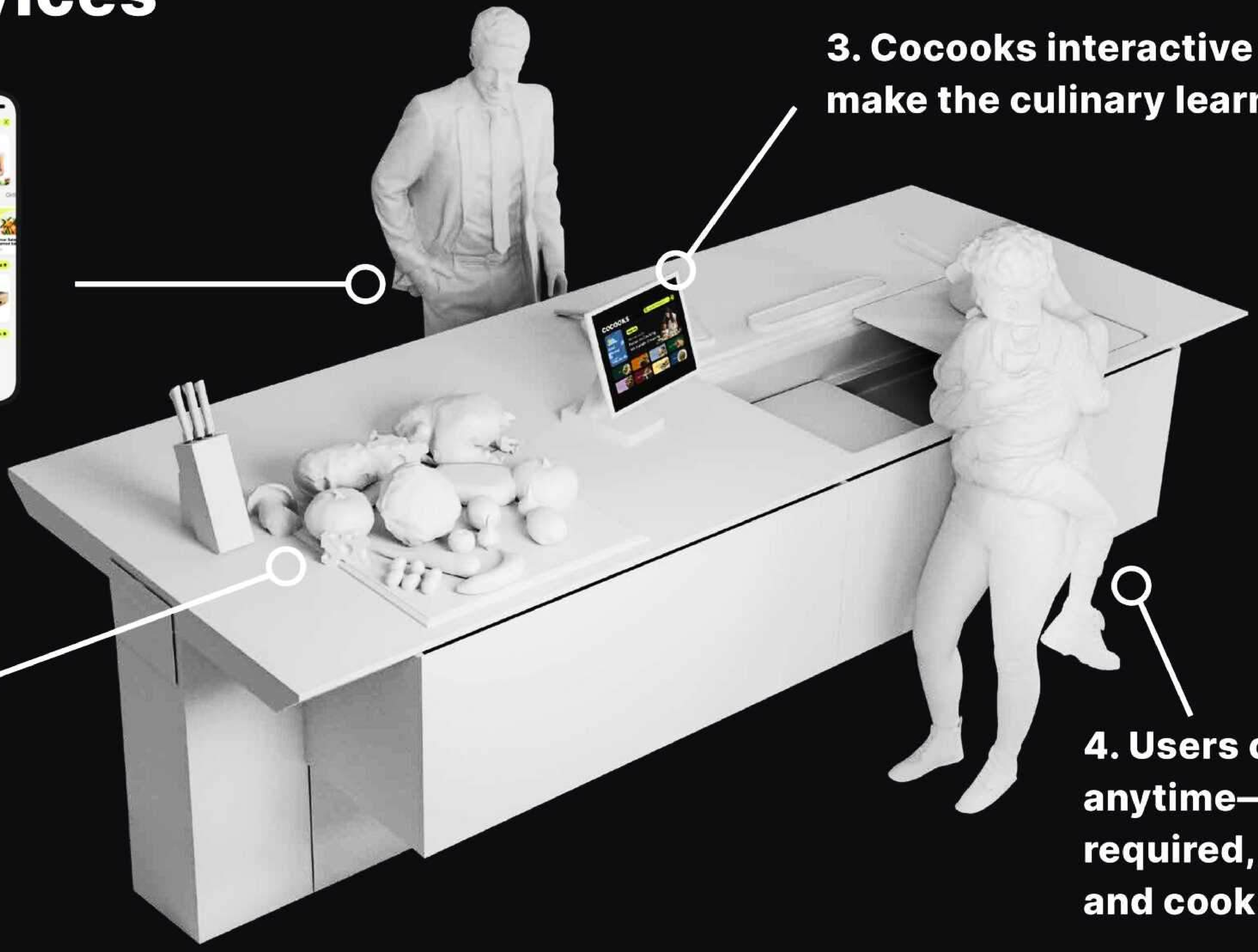
Designed a holistic ecosystem bridging the digital and physical realms. By integrating an intuitive mobile app with shared DIY cooking spaces, Cocooks empowers young adults to seamlessly embrace healthy and affordable meal preparation.

PROJECT TYPE
UX/UI DESIGN

Cocooks, An Affordable DIY Cooking Solution That Integrates Services

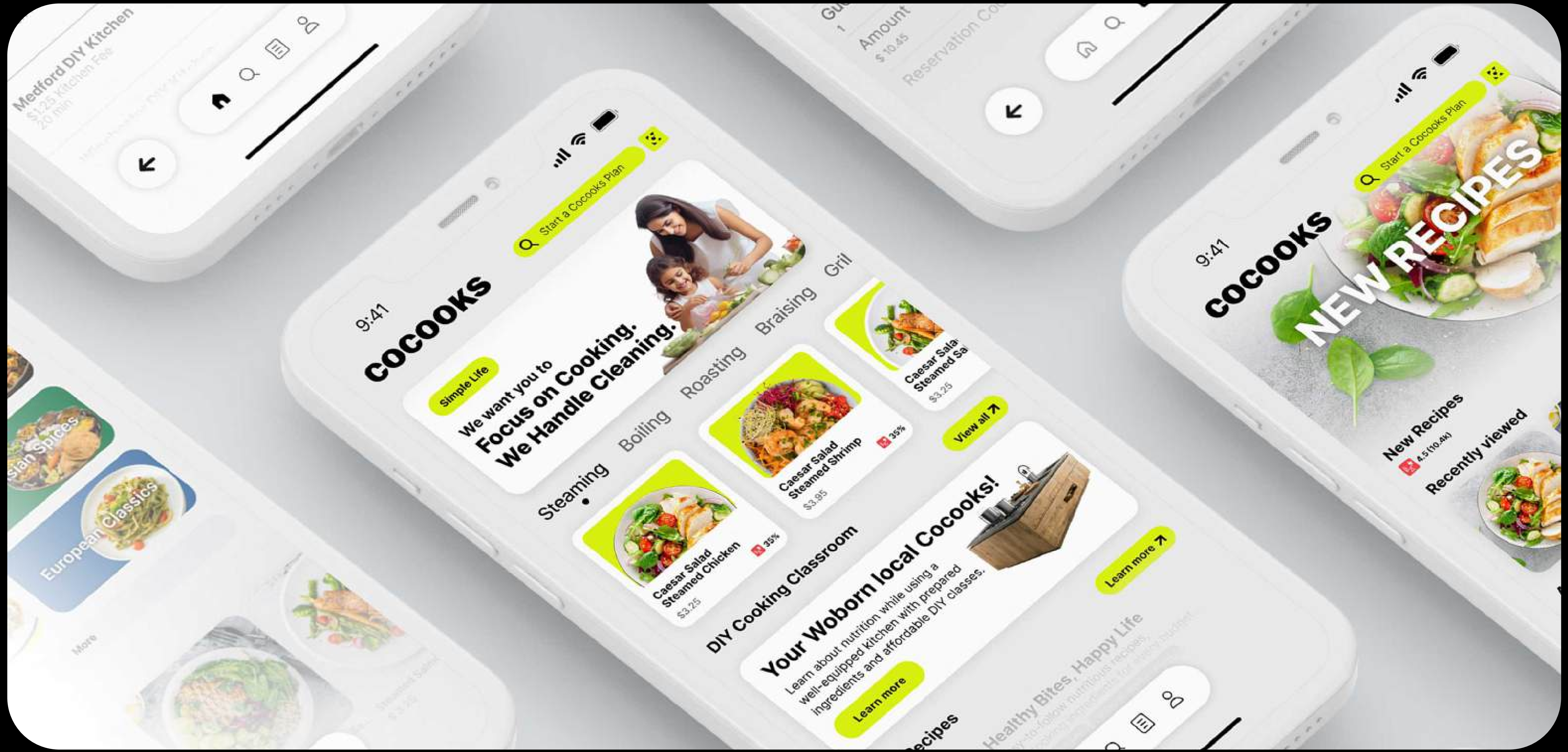
1. Foodies uses our mobile app to book cooking classes.

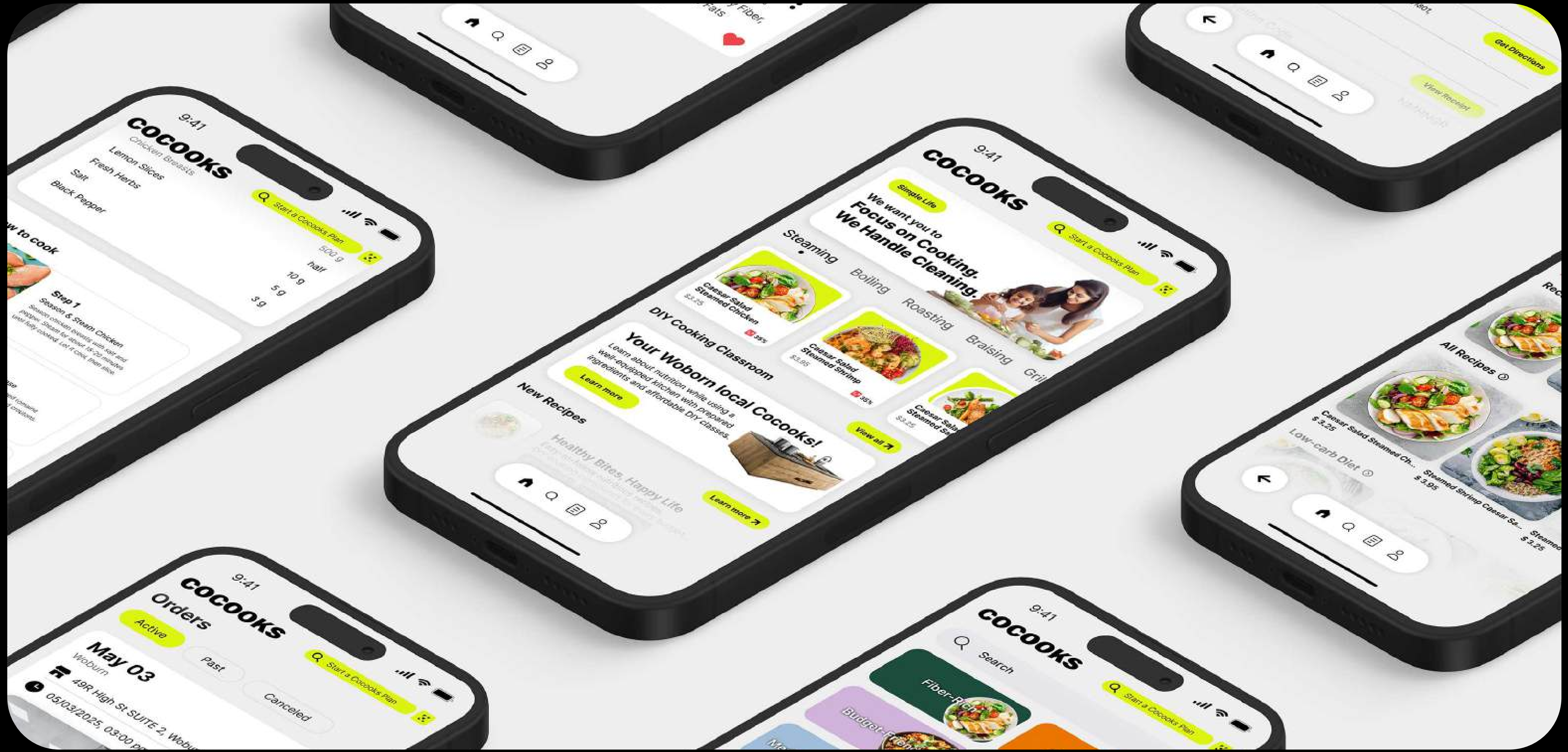
3. Cocooks interactive tutorials make the culinary learnings easy.

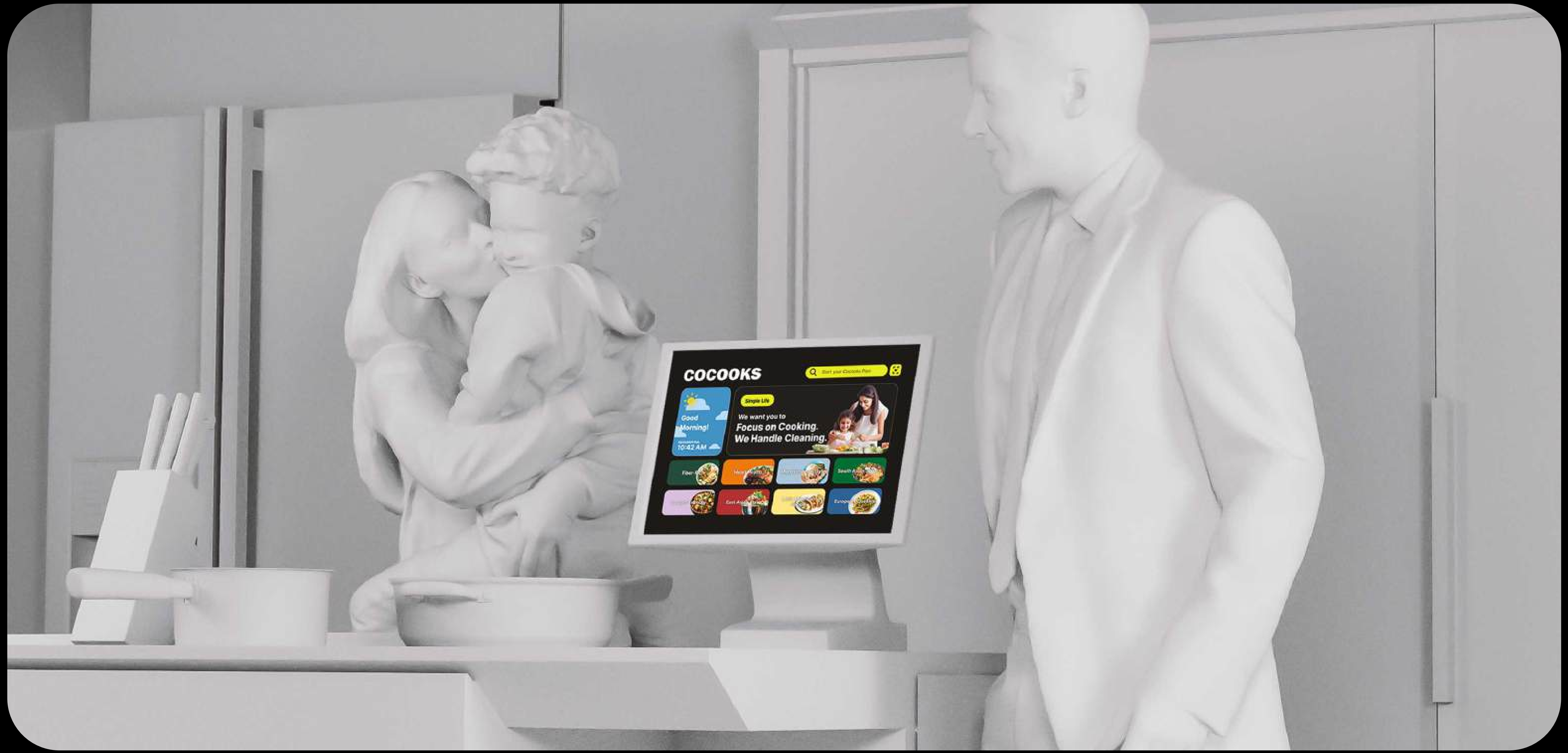


2. Cocooks provide space and cooking kits.

4. Users can drop in anytime—no planning required, just come and cook right away!


9:41
COCOOKS
Start a Cocooks Plan
Simple Life
We want you to
Focus on Cooking.
We Handle Cleaning.
Steaming
Boiling
Roasting
Braising
Caesar Salad Steamed Chicken
Caesar Salad Steamed Shrimp
View all
DIY Cooking Classroom
Your Woborn local Cocooks!
Learn more
NEW RECIPES
New Recipes
Recently viewed


COCOOKS
We want you to
Focus on Cooking.
We Handle Cleaning.
DIY Cooking Classroom
Your Woborn local Cocooks!
New Recipes
Orders
Active
Past
Canceled
May 03


COCOOKS


COCOOKS
Start your Cocooks Plan
Good Morning!
10:42 AM
Simple Life
We want you to
Focus on Cooking.
We Handle Cleaning.
Fiber-Rich
Heart Health
Mediterranean Style
South Asian Spices
Budget-Friendly
East Asian Style
European Classics

mit
media
lab

- Bridging creative strategy with advanced technology at the MIT Media Lab, I engaged in diverse HCI research.

PROJECT SCOPE
PRODUCT DESIGN, USER TESTING, PROTOTYPING

mit media lab
Robotic Limb Project

We conducted 22 user tests to evaluate AR-robotic limb interactions. By identifying critical pain points through real-time feedback, we guided optimization for the next prototyping phase.

PROJECT TYPE
UX RESEARCH / PROTOTYPING

FUTURE X FUTURE

MIT Robotic Limb Case Study

AR Controlled Continuum Supernumerary Robotic Limb (ARCCSRL)

Alice Cai, Aida Baradari, Shea Landeene, Harry Liao, Deepika Gopalakrishnan, Tushar Kanade

Abstract

Supernumerary robotic limbs (SRLs) have the potential to improve human abilities by extending the range and number of actions that can be performed. However, the existing control interfaces for SRLs are low-level, requiring a high cognitive load for users. This research proposes a novel augmented reality (AR)-controlled continuum supernumerary robotic limb (ARCCSRL) that allows users to deliver high-level control commands using eye gaze input supported by secondary voice commands. To assess the effectiveness, usability, and limitations of this system, comparative user studies were conducted using a joystick-controlled continuum SRL and an AR-controlled continuum SRL. Given several technical difficulties during user testing, the results do not suggest that using AR as a control method leads to improved accuracy or efficiency in human-robot interaction, but does leave room for improvements in future studies to further validate the hypothesis of AR controls serving to improve accuracy and efficiency compared to traditional control mechanisms.

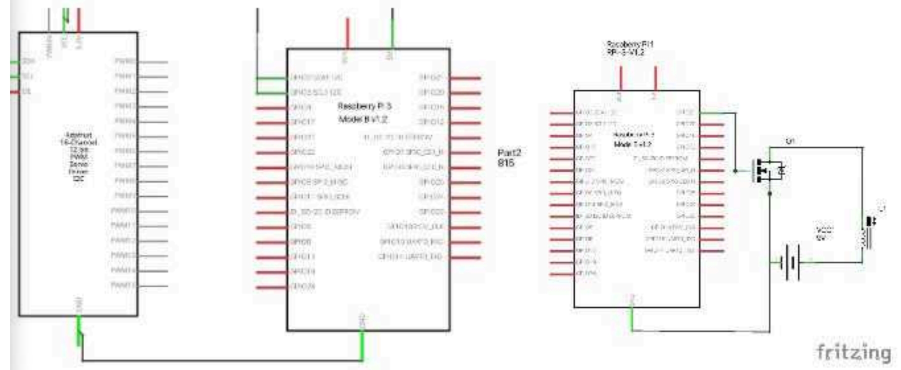
Keywords— supernumerary robotic limb, continuum robot, augmented reality, eye-gaze tracking, voice command recognition, embodiment

I. Introduction

Supernumerary robotic limbs (SRLs) are robotic appendages worn by human users [1]. SRLs have the potential to improve human abilities by extending the range and quantity of actions that can be performed, especially for those with disabilities [2]. However, the current control interfaces for SRLs are often low-level, which results in both a steep learning curve and a high cognitive load for the human user [1]. This can limit the effectiveness and usability of these devices. As such, there is a need for a more natural, intuitive, and user-friendly control system. Additionally, most SRLs are rigid robots designed similarly to human limbs. While this provides a familiar form factor, it limits the possible range of motions, applications, and uses for these devices. To improve the usability and effectiveness of SRLs, there is a need to explore alternative designs and control systems that leverage the potential of the technology to create more versatile and user-friendly limbs.

This research proposes a novel AR-controlled continuum supernumerary robotic limb (ARCCSRL) that allows users to deliver high-level control commands using eye gaze input supported by secondary voice commands. Continuum robots have shown greater flexibility and adaptability compared to rigid robots, making them suitable for integrating with the human body for a wider range of applications and environments [3]. In this phase of research, the control system focuses on allowing users to direct the SRL to move toward and grab objects in the operation space. This research contributes (1) a three-segment continuum SRL hardware design, (2) a machine learning-based proportional-integral-derivative (PID) robotic controller for the robotic limb, (3) a novel AR control language using eye gaze and voice command to identify target objects and desired operations, and (4) insights from human subject experimentation using the SRL. By using high-level commands and taking advantage of the computational capabilities of the machine in the human-machine system, the research aims to reduce the cognitive load and learning curve of controlling the SRL. Additionally, the proposed control system can be extended to off-body machines, allowing for more effective coordination and teamwork between humans and machines. It is hypothesized that compared to traditional control methods such as a joystick, the ARCCSRL is expected to demonstrate enhanced user performance and enhanced intuitive control via shorter average time to complete tasks, greater average accuracy in movement execution, and greater average rating in user experience.

[1] B. Yang, J. Huang, X. Chen, C. Xiong and Y. Hasegawa, "Supernumerary Robotic Limbs: A Review and Future Outlook," in IEEE Transactions on Medical Robotics and Bionics, vol. 3, no. 3, pp. 623-639, Aug. 2021, doi: 10.1109/TMRB.2021.3086016.
[2] Prattichizzo, Domenico, et al. "Human augmentation by wearable supernumerary robotic limbs: review and perspectives." Progress in Biomedical Engineering 3.4 (2021): 042005.
[3] Elena, Giannaccini Maria. "Novel design of a soft lightweight pneumatic continuum robot arm with decoupled variable stiffness and positioning." Soft robotics (2018).



ɛft: Simple servo driver to Raspberry Pi connection, Right: Electromagnet actuation circuit diagram

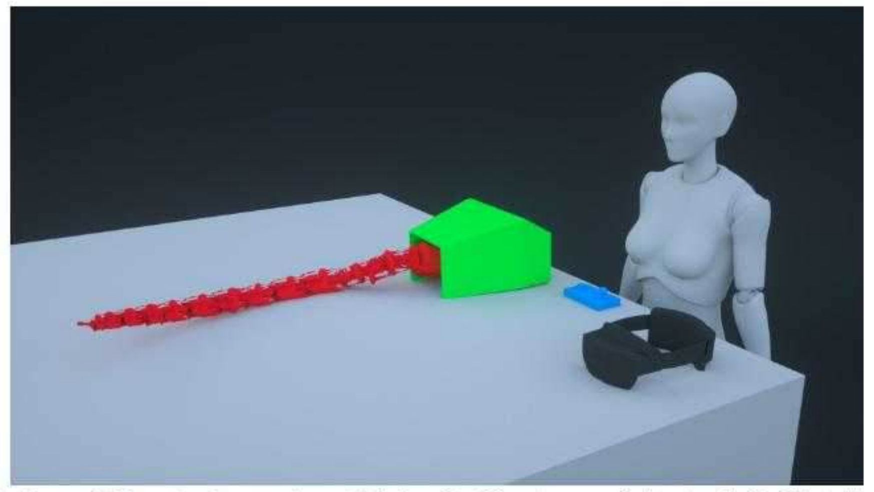
sing a MOSFET and an additional, external 9V power supply, the electromagnet can be directly controlled from the ıspberry Pi.

ɔth of these circuits are used in the manual and AR controller, and are therefore generalizable. The manual controller es both Joystick and buttons to control the servo motors and the electromagnet respectively, whereas the AR ntroller uses predicted positions and language to control servo motors and the electromagnet respectively.

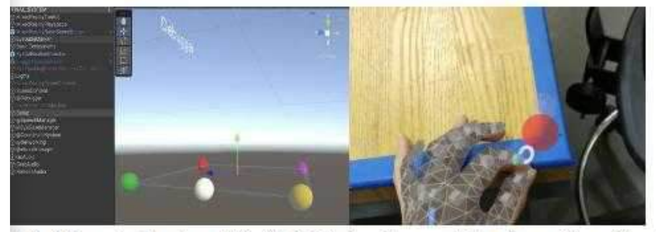
achine Learning Controller. To control the SRL, a machine learning model was developed to map the desired end fector position to the appropriate set of servo angles. Instead of a traditional kinematics-based controller,, we trained ıeural network to predict the needed servo angles for a desired location. We developed a data collection rig consisting an overhead camera to collect video data of the limb configurations at different combinations of servo angles. We veeped a large parameter space, with over 800 data points and developed a computer-vision based post-processing stem to level the alignment, segment the joints and end effectors, and map the pixel positions to real world mensions.

The set of collected data shows an elliptic configuration space which we will be trying to predict with our neural network.

Experimental Design



g 1. Continuum SRL user testing experimental design: *Participants were asked to stand behind the table-mounted ntinuum SRL and use either a joystick controller or AR controller to execute the task of moving the SRL end effector the center of a metal target, and activate the electromagnet at the tip of the SRL end effector to grab the metal target; ırticipants were allotted 5 minutes of training with each controller system in which they could engage with the spective controller system freely and ask clarifying questions; Three trials were conducted for each controller type; ial 1 included a metal target to the left of the SRL, Trial 2 included a metal target to the right of the SRL, Trial 3 cluded a metal target in the center in front of the SRL; Participants completed the series of trials in the same order rail 1-3), using the ending location of the previous trial as the origin point for the next trial; Accuracy was measured ı the distance between the center of the SRL end effector and the center of the metal target; Efficiency was measured ı the time to execute each task*



g 2. AR control system. *(Right) Unity development interface with calibration system. (Left) Researcher calibrating stem by moving spherical table corner markers.*

roller pe	Trial	Avg. time to target (sec)	Avg. distance from target center (mm)	Total trials	Participants per trial (N)
AR	Trial1 (left)	21.8	21.3	20	20
	Trial2 (right)	15.8	28.7	17	17
	Trial3 (center)	15.5	28.8	18	18
	Overall	***17.9***	***26.1***	***55***	***21***
oystick	Trial1 (left)	20.8	12.1	22	22
	Trial2 (right)	22.6	14.0	22	22
	Trial3 (center)	15.4	10.4	22	22
	Overall	***19.6***	***12.2***	***66***	***22***

Continuum SRL user testing results, summary ($N_{AR\text{-}controller}$=21; $N_{joystick\text{-}controller}$=22): *Comparison of AR ed SRL and joystick controlled SRL when completing three trials in which the user was instructed to use the ve controller system to move the SRL end effector to touch the center of a metal target and activate the ıagnet at the tip of the SRL end effector; Green denotes statistically significant outperformance; Red denotes ally significant underperformance; Grey denotes not statistically significant difference*

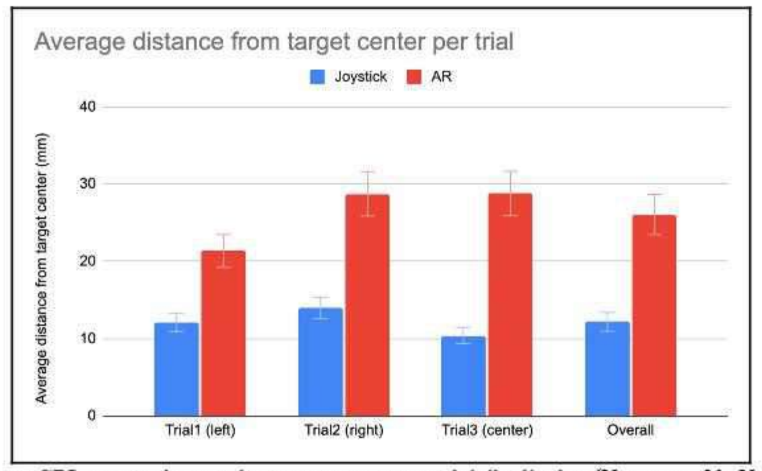
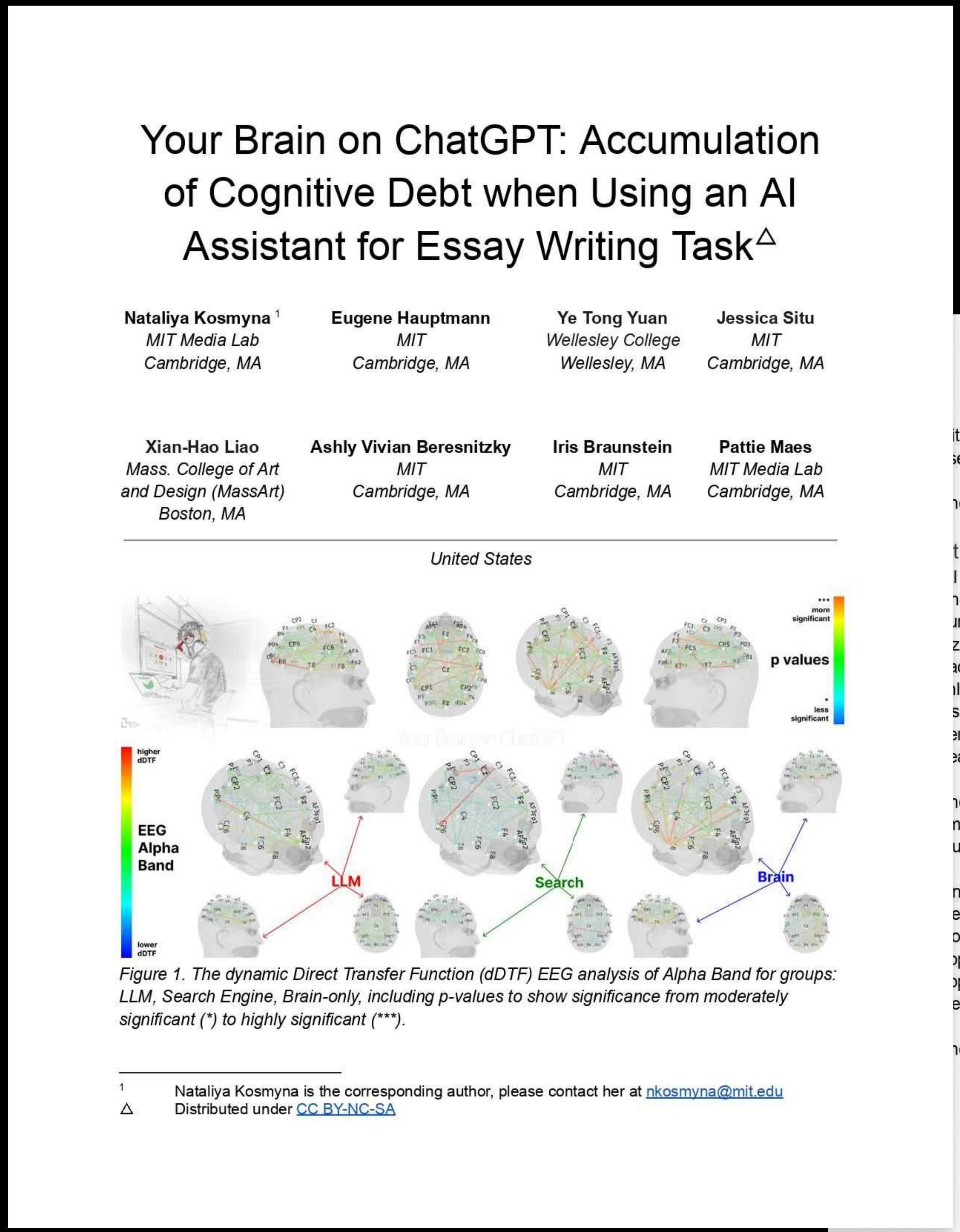


Fig 2. Continuum SRL user testing results, average accuracy trial distribution ($N_{AR\text{-}controller}$=21; $N_{joystick\text{-}controller}$=22):

MIT Brain+AI Case Study



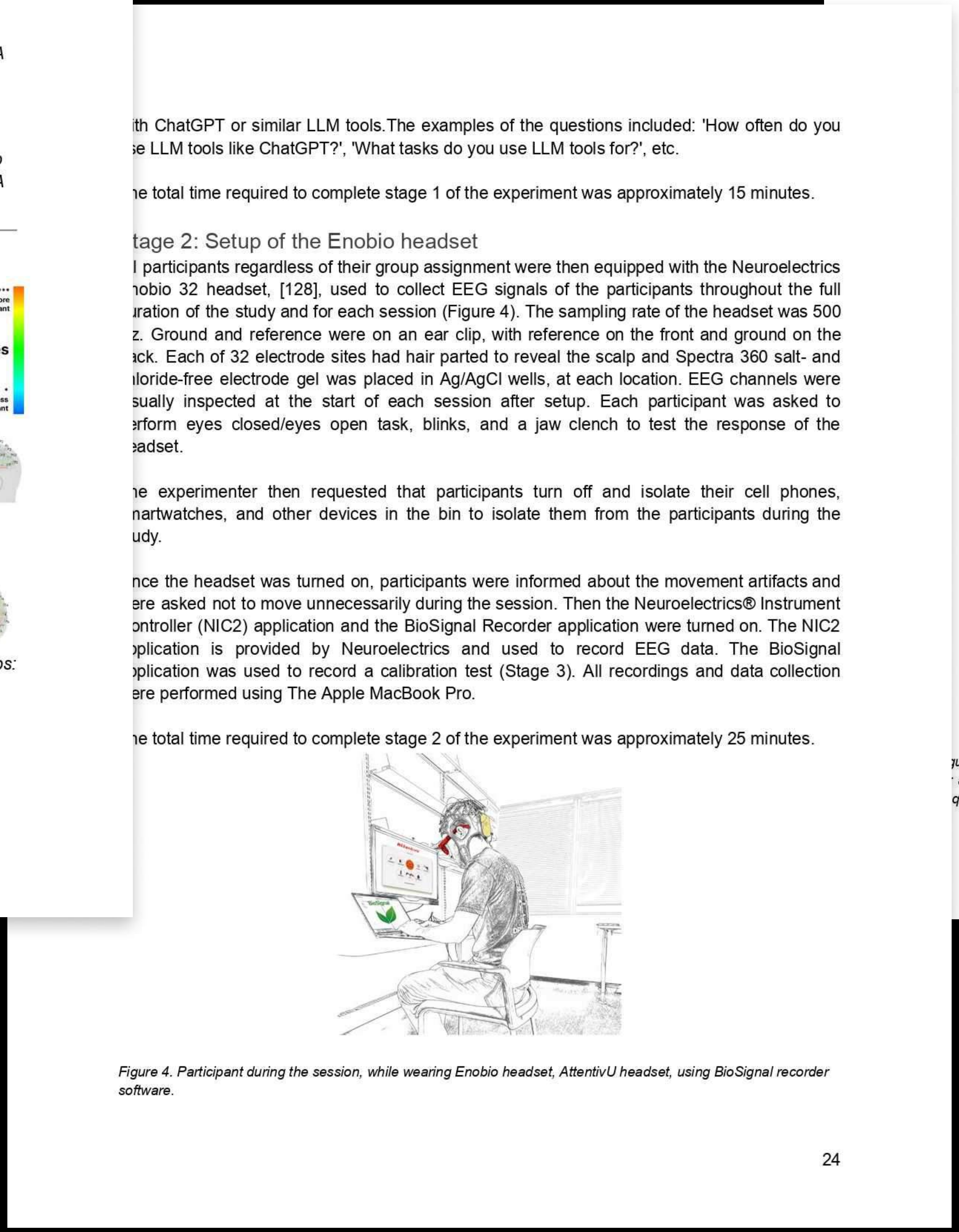
Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task△

Nataliya Kosmyna [1] — MIT Media Lab, Cambridge, MA
Eugene Hauptmann — MIT, Cambridge, MA
Ye Tong Yuan — Wellesley College, Wellesley, MA
Jessica Situ — MIT, Cambridge, MA
Xian-Hao Liao — Mass. College of Art and Design (MassArt), Boston, MA
Ashly Vivian Beresnitzky — MIT, Cambridge, MA
Iris Braunstein — MIT, Cambridge, MA
Pattie Maes — MIT Media Lab, Cambridge, MA

United States

Figure 1. The dynamic Direct Transfer Function (dDTF) EEG analysis of Alpha Band for groups: LLM, Search Engine, Brain-only, including p-values to show significance from moderately significant (*) to highly significant (***).

[1] Nataliya Kosmyna is the corresponding author, please contact her at nkosmyna@mit.edu



ith ChatGPT or similar LLM tools.The examples of the questions included: 'How often do you se LLM tools like ChatGPT?', 'What tasks do you use LLM tools for?', etc.

he total time required to complete stage 1 of the experiment was approximately 15 minutes.

tage 2: Setup of the Enobio headset

l participants regardless of their group assignment were then equipped with the Neuroelectrics nobio 32 headset, [128], used to collect EEG signals of the participants throughout the full uration of the study and for each session (Figure 4). The sampling rate of the headset was 500 z. Ground and reference were on an ear clip, with reference on the front and ground on the ack. Each of 32 electrode sites had hair parted to reveal the scalp and Spectra 360 salt- and hloride-free electrode gel was placed in Ag/AgCl wells, at each location. EEG channels were isually inspected at the start of each session after setup. Each participant was asked to erform eyes closed/eyes open task, blinks, and a jaw clench to test the response of the eadset.

he experimenter then requested that participants turn off and isolate their cell phones, martwatches, and other devices in the bin to isolate them from the participants during the udy.

nce the headset was turned on, participants were informed about the movement artifacts and ere asked not to move unnecessarily during the session. Then the Neuroelectrics® Instrument ontroller (NIC2) application and the BioSignal Recorder application were turned on. The NIC2 pplication is provided by Neuroelectrics and used to record EEG data. The BioSignal pplication was used to record a calibration test (Stage 3). All recordings and data collection ere performed using The Apple MacBook Pro.

he total time required to complete stage 2 of the experiment was approximately 25 minutes.

Figure 4. Participant during the session, while wearing Enobio headset, AttentivU headset, using BioSignal recorder software.

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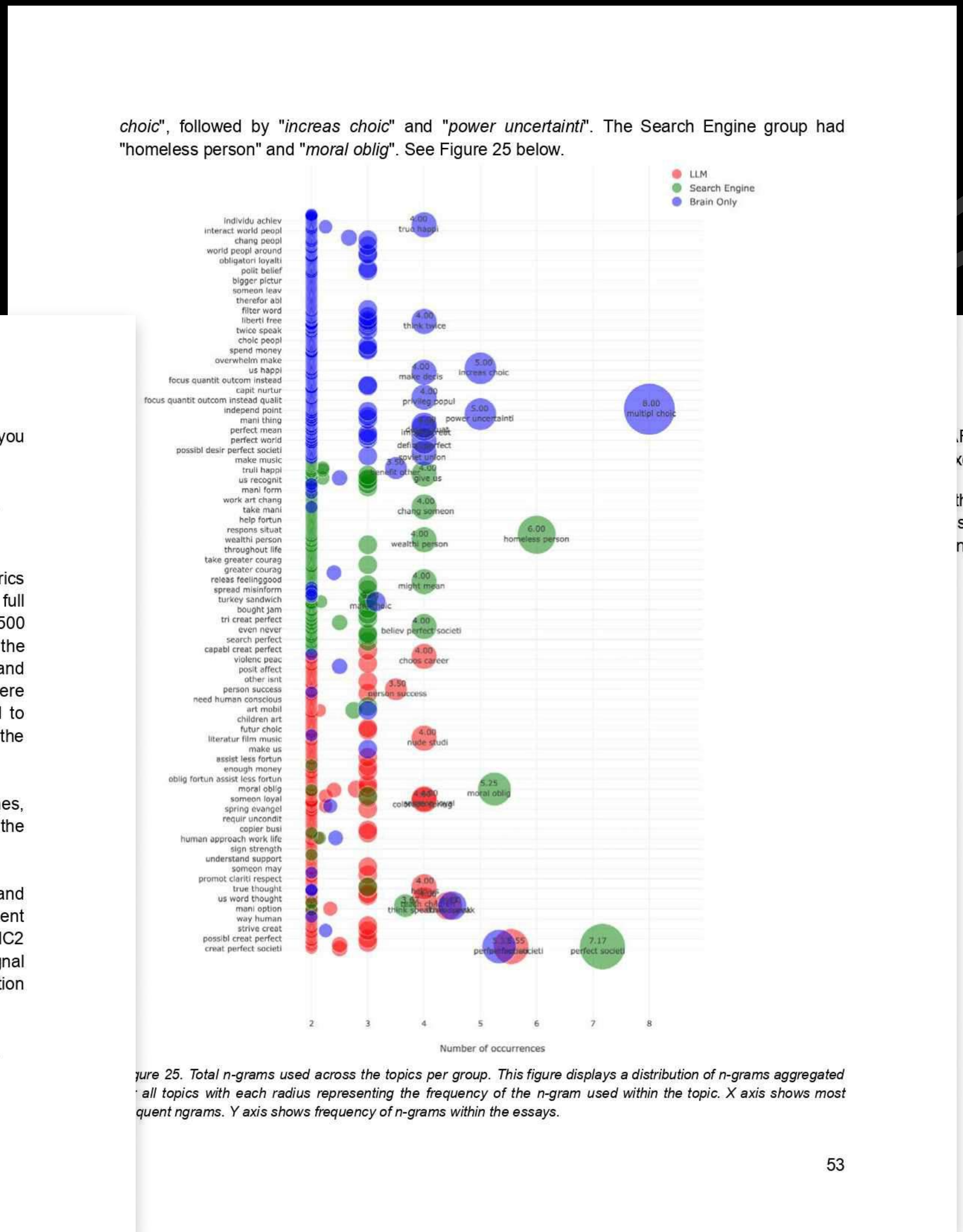
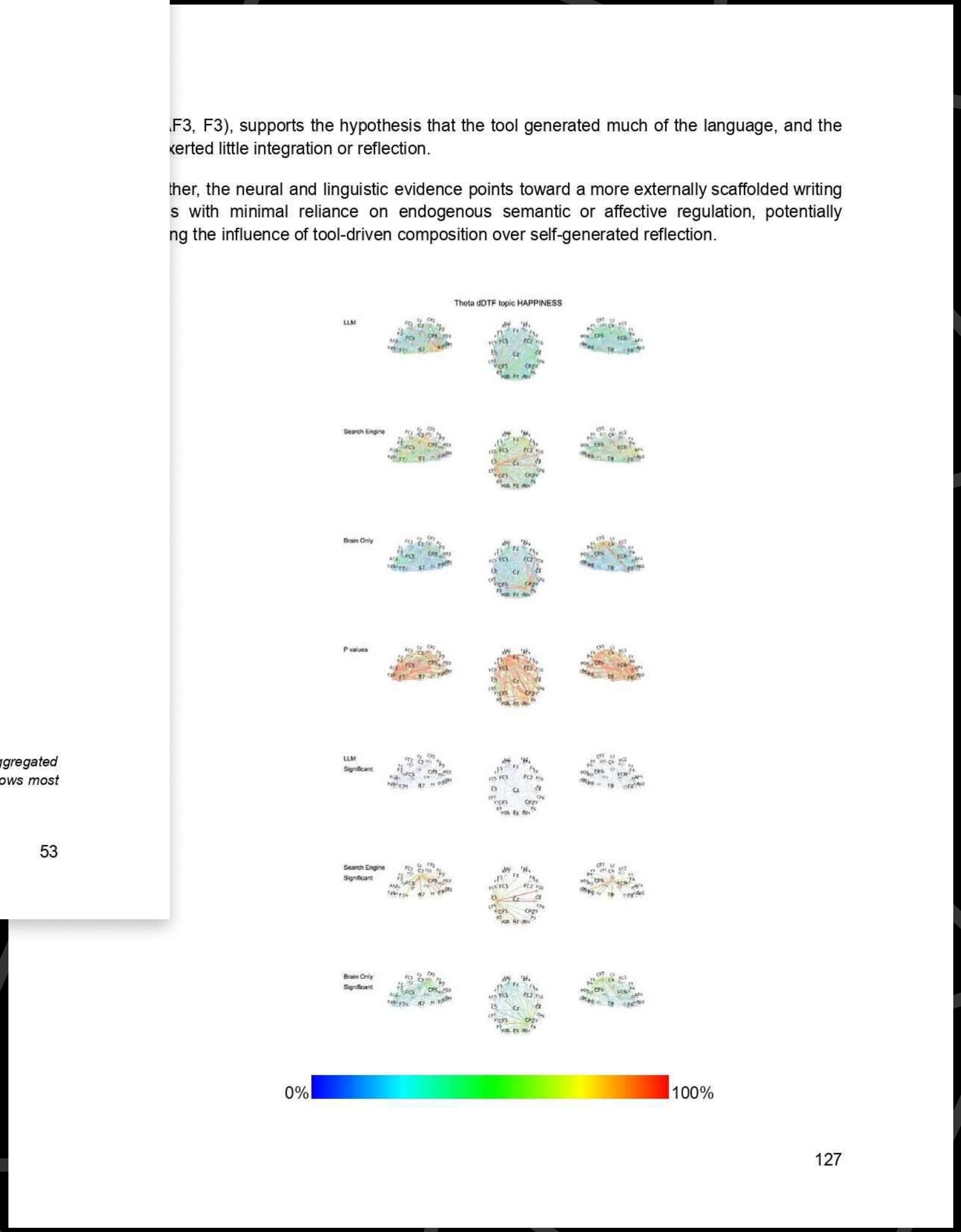


choic", followed by "*increas choic*" and "*power uncertainti*". The Search Engine group had "homeless person" and "*moral oblig*". See Figure 25 below.

gure 25. Total n-grams used across the topics per group. This figure displays a distribution of n-grams aggregated all topics with each radius representing the frequency of the n-gram used within the topic. X axis shows most quent ngrams. Y axis shows frequency of n-grams within the essays.

53



F3, F3), supports the hypothesis that the tool generated much of the language, and the xerted little integration or reflection.

ther, the neural and linguistic evidence points toward a more externally scaffolded writing s with minimal reliance on endogenous semantic or affective regulation, potentially ng the influence of tool-driven composition over self-generated reflection.

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DREAM REALITY

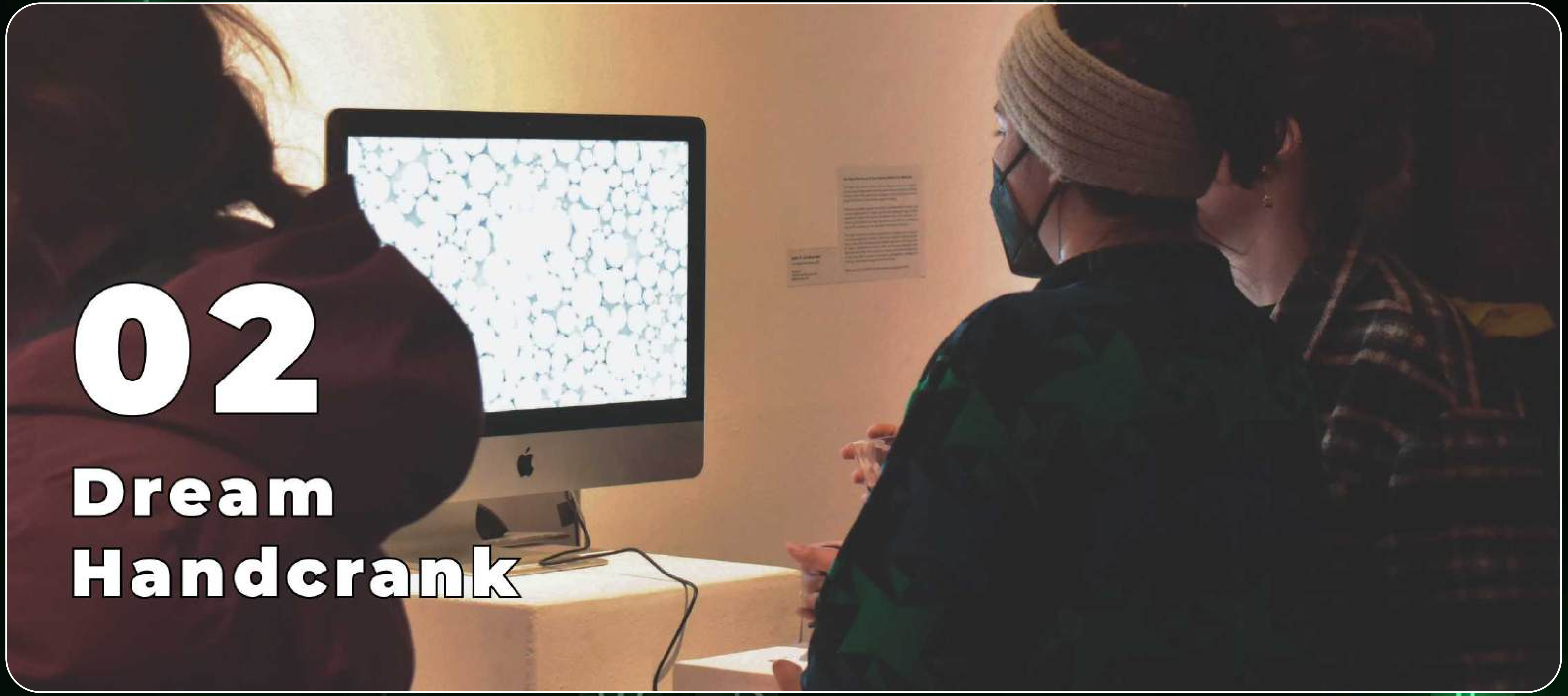
- Investigated the intersection of dream, memory, and perception through an immersive master's thesis, employing deep user research and an interactive hardware system.

PROJECT SCOPE
THESIS RESEARCH, EXPERIENCE DESIGN, HARDWARE PROTOTYPING


Lucid Dreams
History Exhibition
Lucid Dreams History Exhibition
Case Study One
Common Threads
IT IS TIME
TO DREAM
Code


DREAM
REALITY
XIAN-HAO (HARRY) LIAO

01
Lucid Dreams History Exhibition

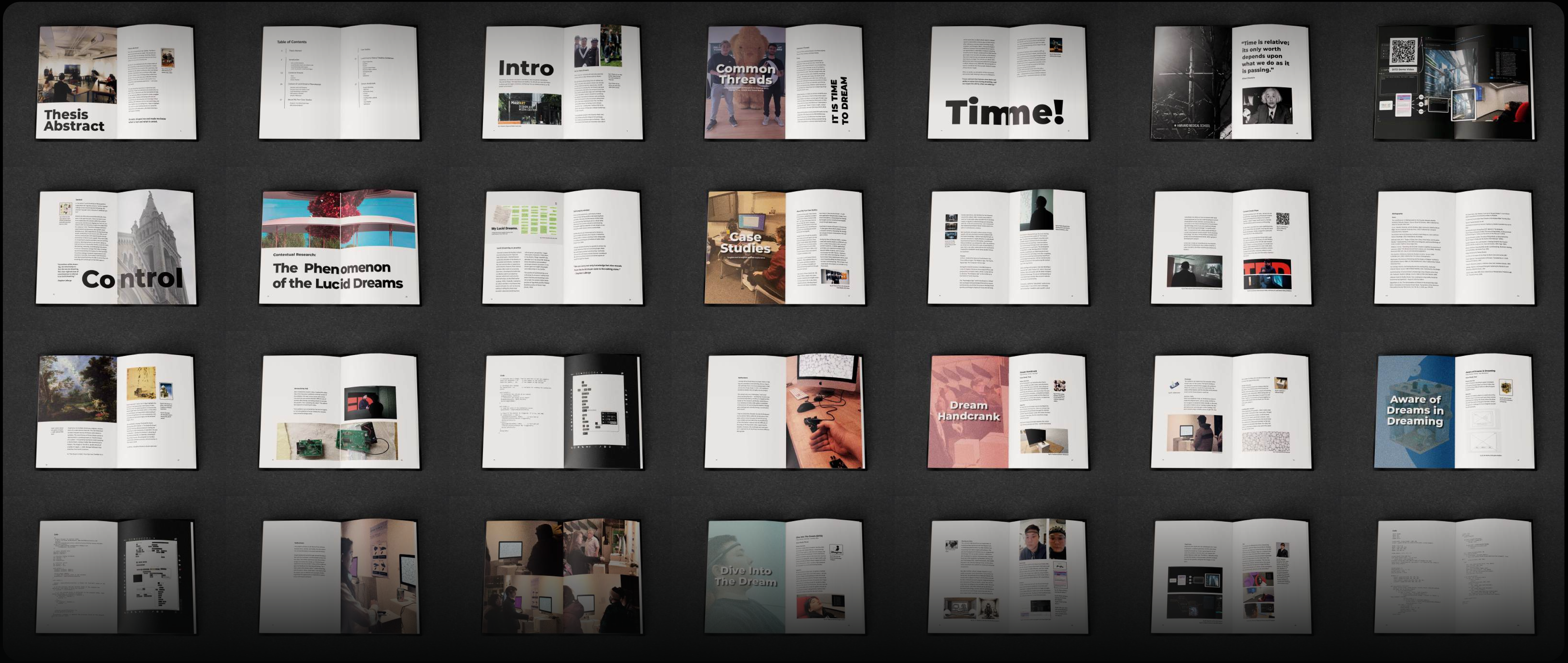


02
Dream Handcrank

03
Dive into the Dream

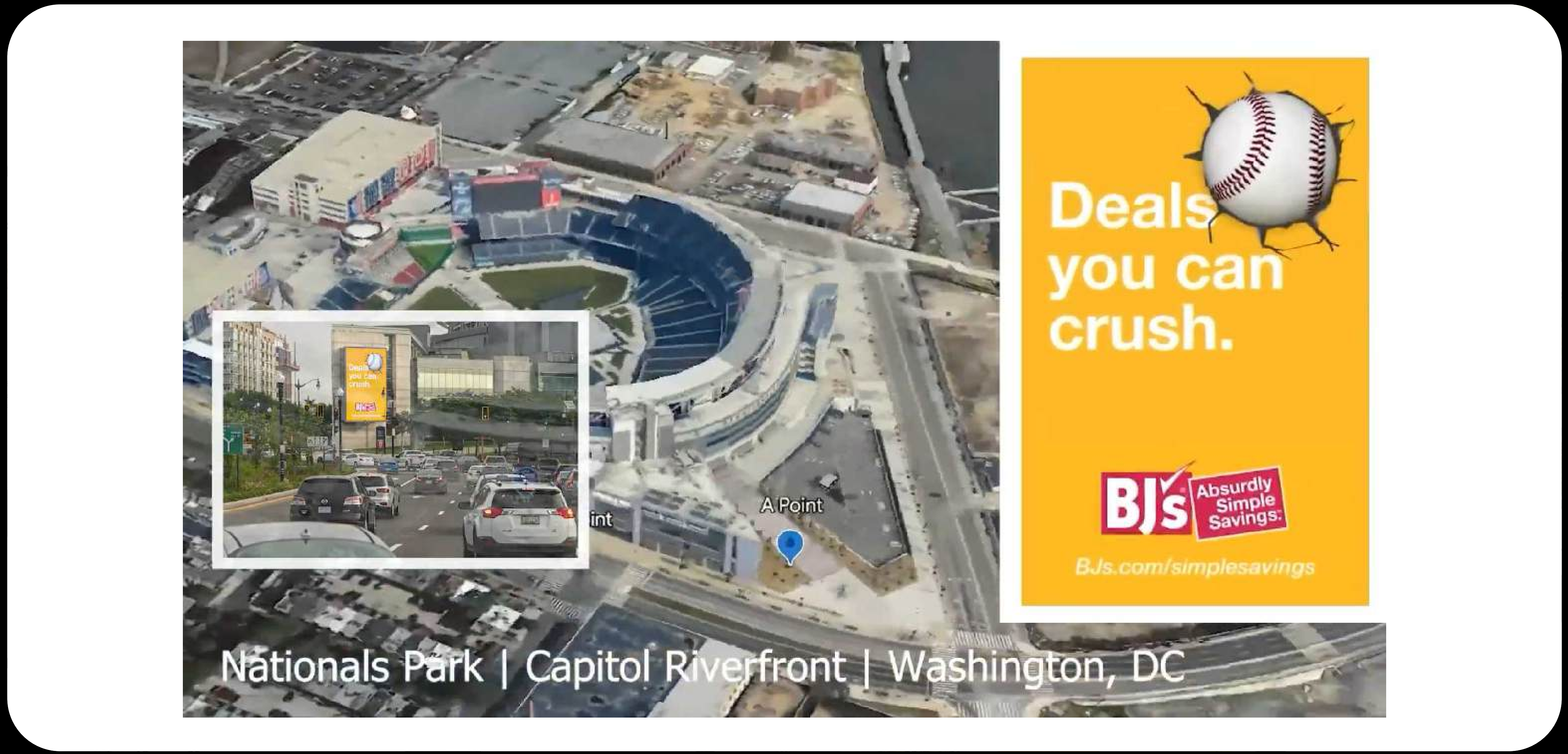
04
Aware of Dreams in Dreaming

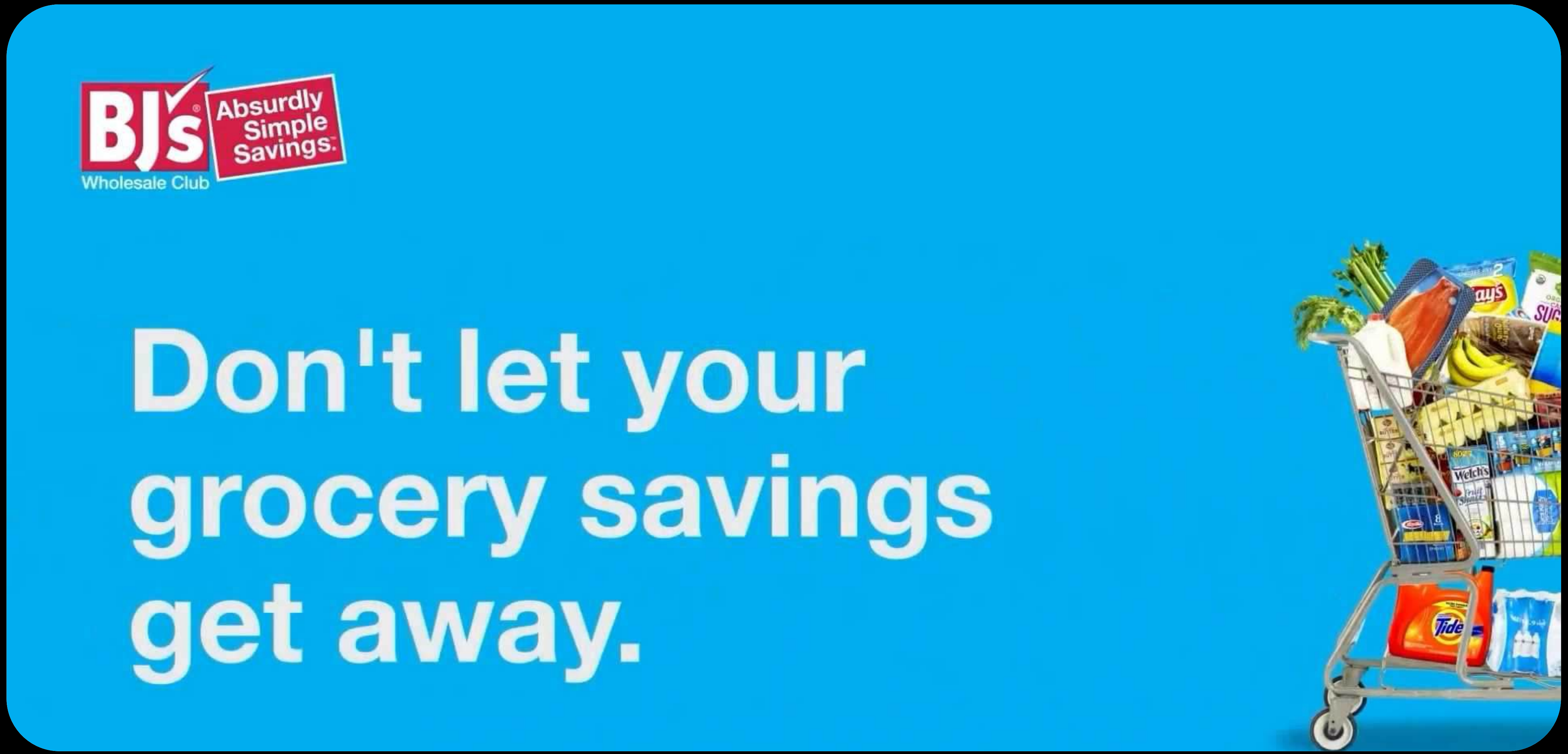
Thesis Book Design

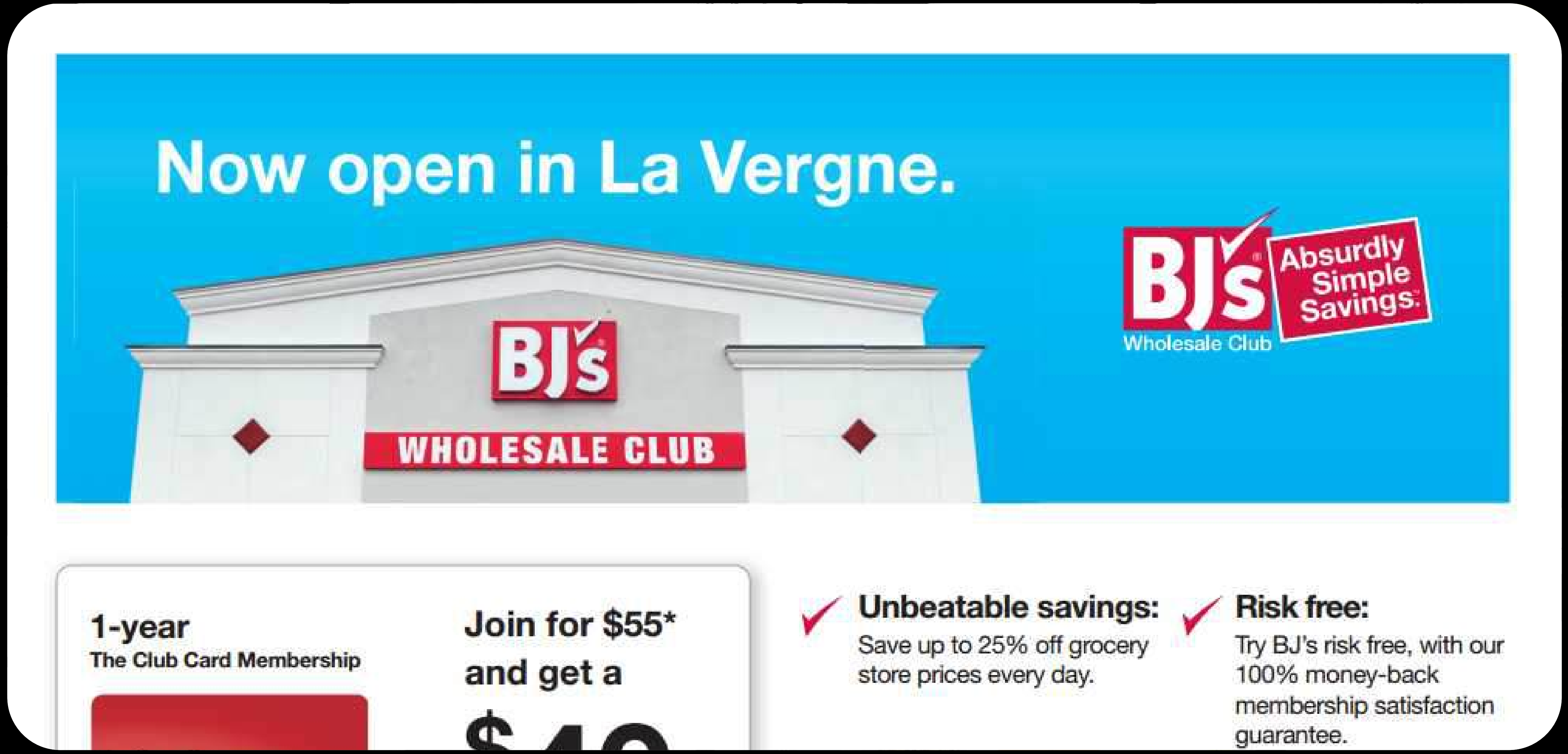


- Led the large-scale visual design for the BJ's Nationals Park OOH Ads campaign in Washington DC during the summer internship.

PROJECT SCOPE
OOH CAMPAIGN, MARKETING MATERIAL DESIGN


Deals you can crush.
BJ's Absurdly Simple Savings.
BJs.com/simplesavings
A Point
Nationals Park | Capitol Riverfront | Washington, DC


BJ's Absurdly Simple Savings.
Wholesale Club
Don't let your grocery savings get away.


Now open in La Vergne.
BJ's
WHOLESALE CLUB
BJ's Absurdly Simple Savings.
Wholesale Club
1-year
The Club Card Membership
Join for $55*
and get a
Unbeatable savings:
Save up to 25% off grocery store prices every day.
Risk free:
Try BJ's risk free, with our 100% money-back membership satisfaction guarantee.


BJ's Absurdly Simple Savings.
Wholesale Club
MDP Summer Internship Final
Harry Liao / MDP Summer Intern - Creative Team
Master of Fine Arts: Dynamic Media Institute / MassArt

DYNAMIC MEDIA INSTITUTE

- The project focused on modernizing the MassArt DMI's digital presence through a contemporary visual language and improved information architecture.

PROJECT SCOPE
UI/UX DESIGN / WEBSITE REDESIGN / BRANDING

DMI Institute Website

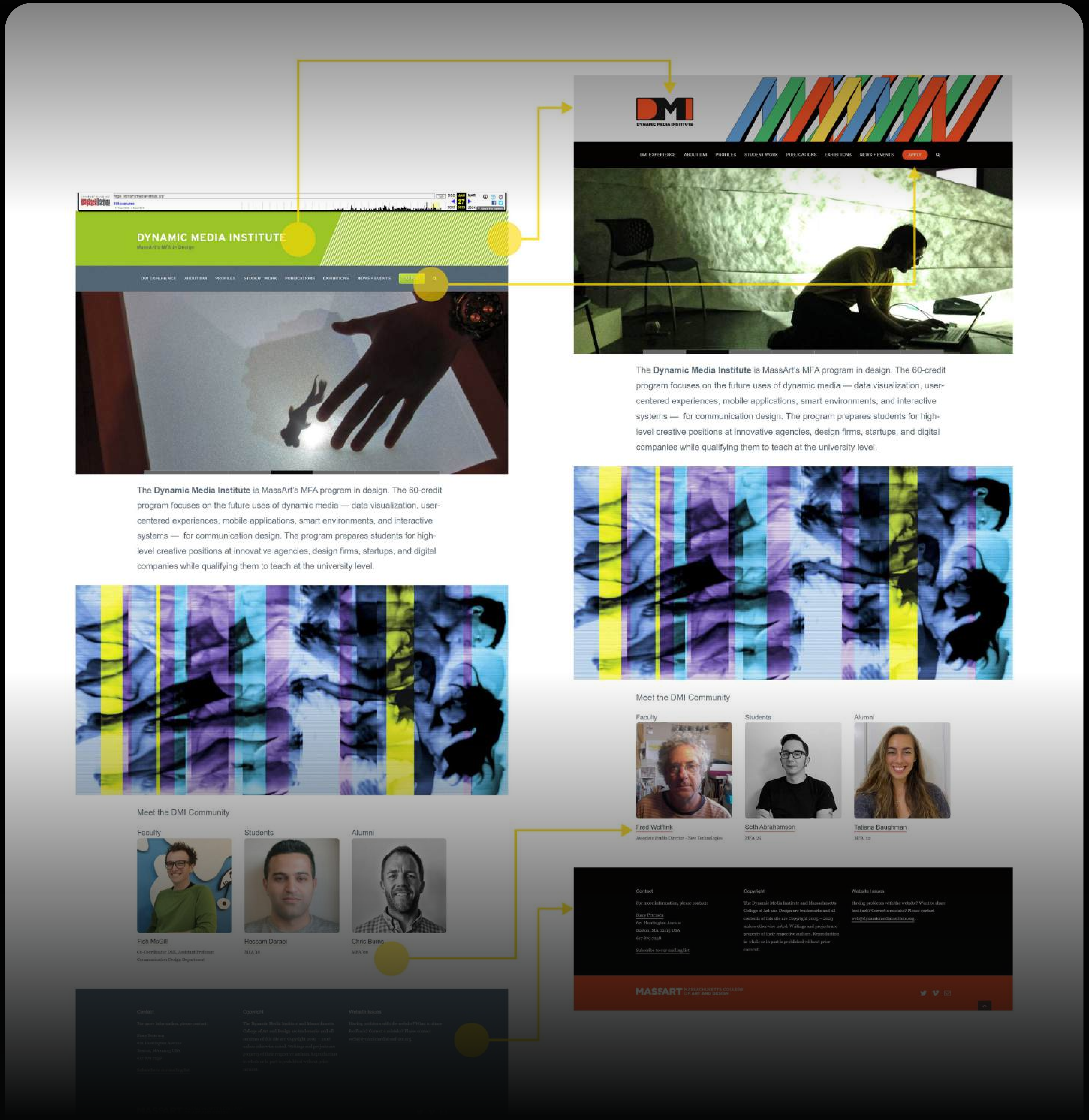
Website Redesign

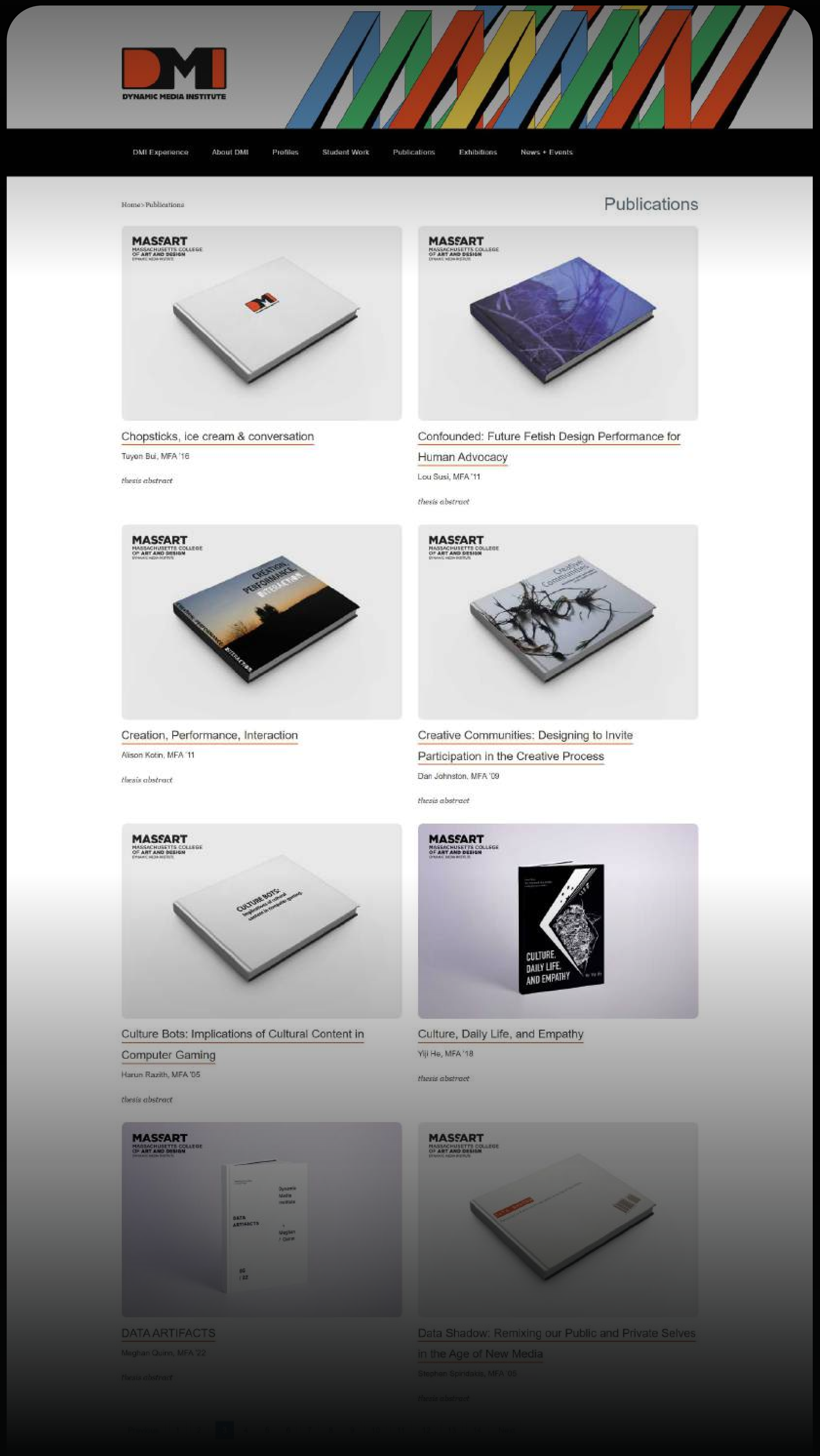
Role: Designer / Developer
Industry: Education
Date: 2024

Institute Website Upgrade
The Dynamic Media Institute (DMI) at MassArt required a visual and functional overhaul to reflect its cutting-edge design curriculum.

Focused on improving user navigation and simplifying the back-end interface for faculty and student content management.

Redesigned layout features a bold, grid-based aesthetic that aligns with the institute's modernist brand identity.





PROJECT TYPE
UI/UX DESIGN / WEBSITE REDESIGN / BRANDING


DYNAMIC
MEDIA
INSTITUTE
Dynamic Media Institute

ABC DESIGN

代言人 **王陽明**

- Involved creating cohesive social media graphics, marketing materials, and promotional short videos to elevate brand presence and engage the target audience.

PROJECT SCOPE
VISUAL DESIGN / SOCIAL MEDIA MARKETING / VIDEO EDITING

ABC DESIGN

DESIGN PORTFOLIO

Brand Campaign
Visual Content & Video Editing

Executed visual strategy and content design for the Sunny Wang ambassador campaign.

A sophisticated and modern visual language was designed to communicate the premium positioning of the high-end stroller line.

Cohesive social media graphics and dynamic short-form videos were created to maximize digital impact.

The campaign successfully elevated the brand's premium image and drove significant engagement.

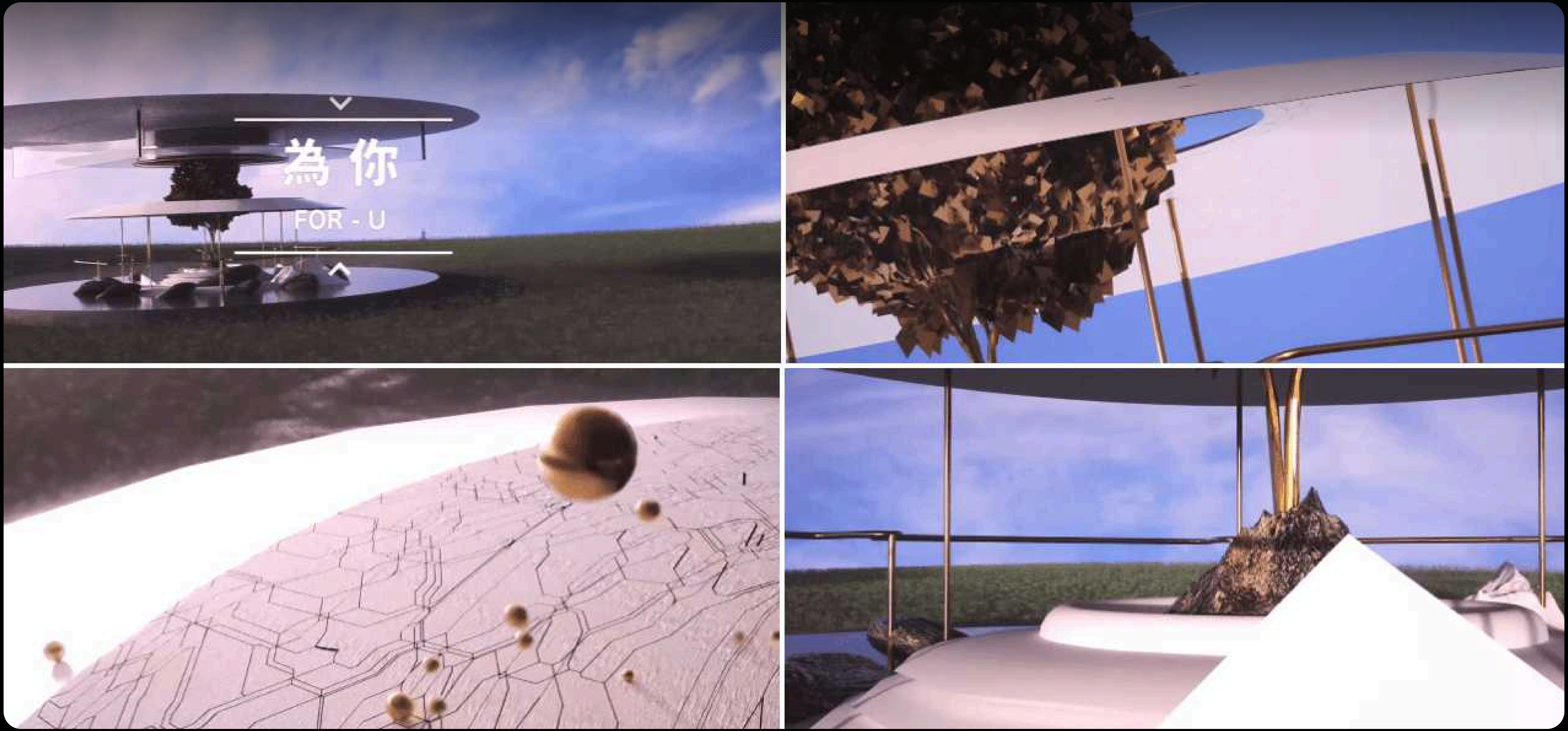
PROJECT TYPE
SOCIAL MEDIA MARKETING


SPORT

FOR-U

- Merging creative strategy with visual storytelling to build a cohesive music video and brand experience.

PROJECT SCOPE
VISUAL STORYTELLING, ART DIRECTION, MV PRODUCING


為你
FOR - U


FOR-U
RENO GORDON
FT. LUNA
A LYRICS VIDEO BY BROTHERHOOD CASUAL
DIRECTED BY RNGDN
3D MODELING.YANG PO ZHI
EDIT&DI.RNGDN
MANAGEMENT. BEAR TIM
LINE PRODUCER
TW
RENO GORDON,XUN XIE
LA
AUSTIN LEEDS
BROTHERHOOD
CASUAL

FOR-U
3D Animation & MV Design

I directed and produced the 3D animated music video, set in a futuristic oasis that merges nature with technology. The project encompassed end-to-end execution, including 3D visual animation, original music production, and comprehensive brand identity design, delivering a cohesive and immersive audiovisual experience.

PROJECT TYPE
3D ANIMATION / MUSIC PRODUCTION / MV DESIGN

C-CUBE
中科大青年創業孵化基地

- An exclusive, integrated platform combining portfolio showcase for student entrepreneurs.

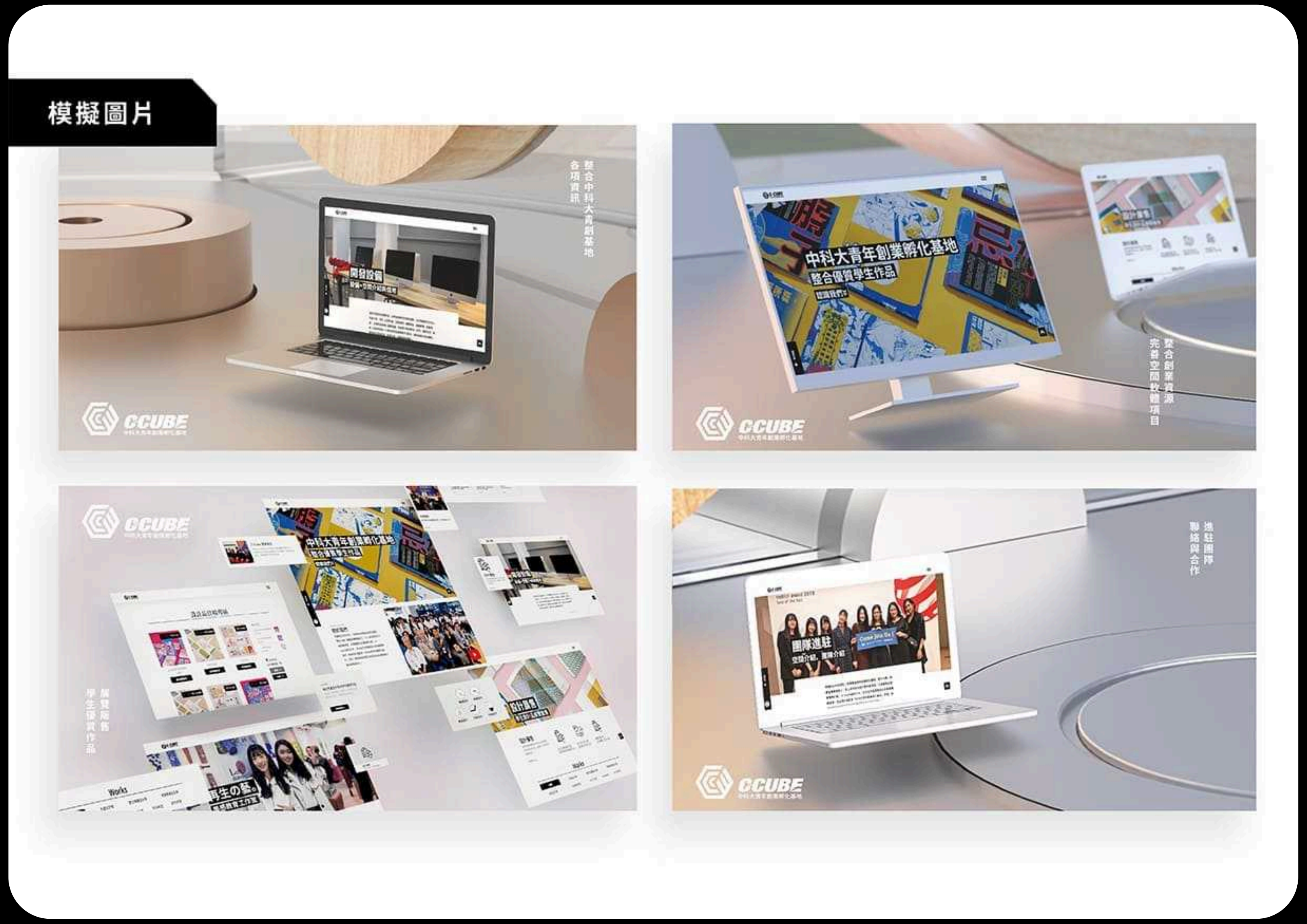
PROJECT SCOPE
UI/UX DESIGN / WEBSITE REDESIGN / BRANDING

C-CUBE
UI/UX Design & Information Architecture

I structured and designed a comprehensive website for the C-CUBE Incubation Base.

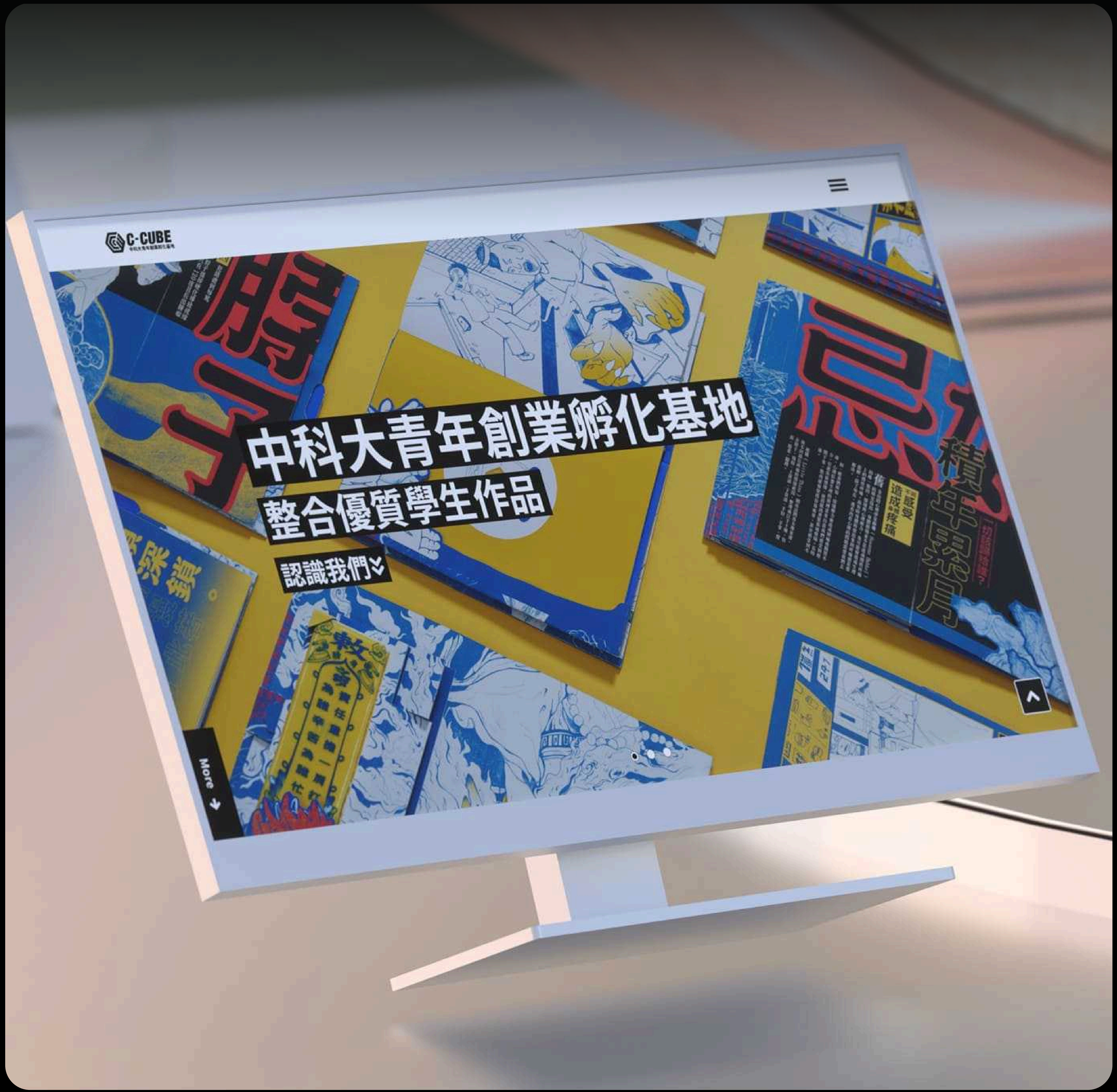
By systematically categorizing their diverse services into six core sections, from entrepreneurship counseling to design exhibitions.

We provides an intuitive and seamless user experience that connects student entrepreneurs with vital resources.



PROJECT TYPE
UI/UX DESIGN / WEB DESIGN


Perspective
Options
Filter
Panel
ProRender
OctaneCamera
phone
Null.1
Null
keyboard
aluminium
touchpad
metal
white_plastic
rubber
screws
display
apple
Mode
Edit
User Data
Basic
Coord.
Object
Physical
Details
Stereoscopic
Composition
Spherical
OctaneCameraTag
Object Properties
Projection
Perspective
Focal Length
100
Custom (mm)
Sensor Size (Film Gate)
32.5
Custom (mm)
35mm Equiv. Focal Length:
110.769 mm
Field of View (Horizontal)
18.46 °
Field of View (Vertical)
13.109 °
Zoom
Film Offset X
0 %
Picture Viewer
File
Edit
View
Compare
Animation
Navigator
Histogram
30 %
History
Info
Layer
Filter
Stereo
History
Name
R
Render Time
F
01poster_final.tif *
00:27:48
0
3507
01poster_final.tif
0
3507
團隊進駐
空間介紹、團隊介紹
Come Join Us !
0 F
210 F
Create
Edit
View
Select
Material
Texture
All
No Layer
Metal


C-CUBE
中科大青年創業孵化基地
整合優質學生作品
認識我們
More


C-CUBE
開發設備
設備+空間介紹與借用
More
這座空間未來發展特色，以特色專業研究室為策略，並分別建構五大核心
發展方向，包含：品牌形象、包裝結構、媒體廣告、圖像敘事、影像應
用！近幾年逐漸購入專業設備，提供學生實務實作、創作、產學合作、教
學、創業等資源，計畫未來將這些設備予以整合，規劃系統化配訓課程，

XIAN-HAO (HARRY) LIAO

THANK YOU

DESIGN PORTFOLIO